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The Association Between Team-Based Model of Care Characteristics and Patient Outcomes in Community Health Centres: A Systematic Review

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ABSTRACT

Background

Community Health Centres (CHCs) are community-governed organizations often located in communities facing significant challenges with social determinants of health. To date, no review has summarized the evidence on the effects of team-based models of care on patient outcomes. The objective of this systematic review was to synthesize the evidence on the characteristics of interprofessional teams practicing in CHCs and their associations with patient outcomes.

Methods

A systematic review was conducted following the guidelines of the Cochrane Handbook for Systematic Reviews of Interventions. In October 2024, an electronic search was performed in five databases and the grey literature using a combination of descriptors and keywords. No restrictions were applied regarding language or year of publication. Screening and data extraction were independently conducted by two reviewers. The included papers focused on interprofessional teams composed of at least three different types of providers (e.g., physician, nurse, and nutritionist) practicing in CHCs and addressing at least one patient outcome (e.g., satisfaction, diabetes management). The risk of bias was assessed by two independent reviewers using the Joanna Briggs Institute checklist.

Results

From 6,309 identified papers, 36 papers were included. Most papers focused on team composition (n=18) and staffing patterns (n=8). Adjustments to staffing

patterns—such as increasing full-time equivalents, expanding team size, and diversifying skill mixes—were associated with improved patient satisfaction and increased number of visits. Adding a chiropractor reduced opioid prescriptions and patient pain, while including a pharmacist improved hepatitis management and patient satisfaction. Inconsistent findings were observed regarding the inclusion of nurse practitioners and physician assistants in interprofessional teams and their impact on the number of patients seen in CHCs. Specialized interprofessional teams addressing diabetes, pain management, and childhood obesity had better health outcomes and care management compared to standard care, highlighting the value of tailored interprofessional collaboration in achieving improved health outcomes for specific populations.

Conclusions

Specific team compositions are associated with improved health outcomes for the populations served by CHCs. Future research is needed to deepen the understanding of the associations between team composition, the type of care provided, and patients' clinical and psychosocial needs.

Keywords

Community health centres, patient outcomes, primary care, systematic review, team-based model of care

BACKGROUND

Community Health Centres (CHCs) are community governed organizations typically located in areas where residents face significant challenges related to social determinants of health and limited access to healthcare. In 2021, Nagel et al. conducted a concept analysis and identified six key attributes that define CHCs: primary care, accessibility, preventive care, a defined population, health promotion, and comprehensive care (Nagel et al., 2021). Firstly, CHCs often serve as the primary point of entry for healthcare among vulnerable populations, including low-income communities, individuals with mental health or substance use issues, immigrants, refugees, and visible minorities. Secondly, the care philosophy of CHCs emphasizes culturally sensitive care, health promotion and prevention, adopting a holistic approach that takes into account both the clinical and psychosocial contexts of patients (Nagel et al., 2021).

Internationally, CHCs are most commonly found in North America and China. In Canada and the U.S. (in the U.S., they are known as Federally Qualified Health Centers, or FQHCs), CHCs share similar governance, organizational structures, and models of care. They are typically governed by community boards with consumer representation and operate as non-profit organizations, funded primarily through government support (Alliance for Healthier Communities, 2020; British Columbia Association of Community Health Centres, 2023; Wright. 2017). CHCs provide a broad range of services, including medical, dental, psychosocial, and educational care, offering services either free of charge or at reduced cost for individuals without insurance or not registered

with a primary care physician. The defining features of CHCs, compared with other primary care settings, are their accessibility to uninsured or unregistered patients, community-based governance, and salary-based clinician compensation (Suleman et al., 2024). In China, CHCs are the main providers of urban primary care, while township hospitals and village clinics serve rural areas. Chinese CHCs operate under three models, government-managed, hospital-managed, or private, and patients are generally expected to pay for services regardless of the model (Li et al., 2015).

To support these purposes, a team-based model of care (TBMC) has been introduced in CHCs since their conception. A TBMC is an evidence-based approach where at least three different types of healthcare professionals (HCPs) collaborate with patients and their families or caregivers to achieve shared treatment goals (Kiran et al., 2020; Kiran et al., 2022; Somé et al., 2020). In 2017, Baik conducted a concept analysis to define and examine team-based care in clinical practice, identifying three core attributes: interprofessional collaboration, a patient-centered approach, and an integrated care process. Current literature agrees on the effectiveness of TBMC for patients, families, interprofessional teams, and healthcare organizations. Compared to traditional physician-focused care, TBMC has shown success in improving health outcomes, enhancing patient experience, reducing healthcare utilization, improving continuity of care, and lowering costs (Kiran et al., 2020; Kiran et al., 2022; Somé et al., 2020).

Several systematic reviews have been published examining the effectiveness of TBMC in various primary healthcare settings (Doose et al., 2022; Hinneh et al., 2024; Li et al., 2023). For instance, a systematic review and meta-analysis by Hinneh et al. (2024) found that TBMC for managing hypertension in Africa led to reductions in blood pressure. Furthermore, their study indicated that the inclusion of specific providers, such as community health workers and nurses, within interprofessional teams contributed to these improvements in blood pressure management (Hinneh et al., 2024). Other reviews have also explored various interprofessional team characteristics and their impact on patient, team, and organizational outcomes. For example, the umbrella review by Wei et al. (2021) suggested that team-level characteristics such as size and adequate staffing can enhance team collaboration, improve care quality, and positively affect patient outcomes (Wei et al., 2022). Most of these reviews encompass a broad range of primary care settings, such as clinics, family health teams, and home-visit programs, yet it remains unclear whether CHCs were included. Considering that CHCs have a distinct governance structure and serve a higher proportion of patients affected by social determinants of health (e.g., low-income individuals, newcomers) compared to other primary care settings (Lewis et al., 2016), this unique context warrants dedicated attention.

Although several literature and systematic reviews have been conducted on primary healthcare, less attention has been paid to synthesizing the literature specifically on TBMC in CHCs, particularly regarding how different team characteristics influence care quality and patient outcomes. Therefore, we argue

that a new systematic review is needed and this is supported by two key points. First, existing reviews, including for example, the systematic reviews published by Marten et al. (2021), Rodis et al. (2022) and Wang et al. (2015) conducted in CHC settings focus on a specific provider or type of care, such as pharmacist-led interventions or telemedicine (Martens et al., 2021; Rodis et al., 2022; Wang et al., 2015). However, prior studies have highlighted the value of interprofessional collaboration across a broader range of providers to improve the accessibility and quality of care for patients and families. Second, while some reviews (Choi et al., 2020; Nakamura et al., 2019) have examined CHC performance and productivity, such as cost and visit volume, less attention has been given to patient- and family-centered outcomes, including patient-reported outcomes, clinical outcomes, and care experiences. To the best of our knowledge, there is no systematic review that assesses the direct benefits of TBMC for patients and families in CHCs.

To address these gaps, we conducted a systematic review aimed at building upon previous work and examining the effects of different characteristics of TBMC in CHCs for communities, patients, and families. Our review specifically focused on TBMC involving at least three different types of providers, consistent with the conceptualization of team-based care proposed by Baik (2017), captures a broad range of TBMC models and characteristics (e.g., team size, composition, dynamics, utilization of e-health, etc.), and assesses patient outcomes. The objectives of this systematic review were to: 1) examine the association between TBMC characteristics and models of care and patient outcomes in CHCs, 2)

identify the team characteristics that significantly impact patient outcomes, and 3) discuss future research directions in this area.

METHODS

Design

A systematic review of the literature was conducted in accordance with the Cochrane Handbook for Systematic Reviews of Interventions guidelines (Higgins et al., 2019). The review protocol was registered in PROSPERO (Audet et al., 2024).

Search Strategy

In October 2024, a search was performed across five databases: 1) MEDLINE, 2) CINAHL, 3) Embase, 4) Cochrane Library, and 5) PsycINFO, to identify papers examining TBMC characteristics and patient outcomes in CHCs. Multiple queries were executed in each database by combining two independent sets of keywords and descriptors: 1) CHCs and 2) TBMC (see Table 1). Additionally, a search of the grey literature was conducted in November 2024, following the Cochrane recommendations (Higgins et al., 2019), to retrieve pre-publication works, reports, and conference proceedings. The grey literature search included: 1) OpenGrey, 2) ClinicalTrials.gov, 3) Google Scholar, 4) National Library of Medicine, and 5) the Virginia Henderson Database. No restrictions on language or publication date were applied. For each included paper, reference lists (backward search) and publications citing the included papers (forward search) were reviewed for potential additional relevant papers. An academic librarian was

consulted to help develop and execute the search strategy. The full search strategies are outlined in Appendix A.

Inclusion and Exclusion Criteria

All published and unpublished quantitative papers addressing TBMC in CHCs were eligible for inclusion. Mixed-method papers were included as well, provided that the quantitative component was presented separately and could be independently extracted from the qualitative portion. Based on Baik's (2017) concept analysis of team-based care, papers were included if they described teams involving interprofessional collaboration among three or more types of healthcare providers and provided direct patient care under a collaborative and integrated approach. All aspects of team characteristics and models of care, such as size, credentials and accreditations, provider types, team dynamics (e.g., relationships, workload), staffing patterns, integration of e-health (e.g., telemedicine, electronic health records), and skill mix, were considered (Baik, 2017). Similarly, based on Liu et al. (2014) concept analysis on patient outcomes, all types of patient outcomes, including clinical and psychosocial outcomes, patient-reported outcome and experience measures (PREMS and PROMs), and healthcare resource utilization, were included (Liu et al., 2014). To be eligible, papers had to focus on one or more team characteristics and their association with one or more patient outcomes. CHCs were defined broadly to include any health center meeting the description provided by Nagel et al. (2021) and were required to meet the following three criteria: 1) be labeled as a CHC, community health clinic, federally qualified health center (FQHC), Centre local de

services communautaires (CLSC), or other related terms; 2) be based in primary healthcare settings; and 3) be publicly funded (Nagel et al., 2021). All types of patient populations were included, and there were no restrictions on geographical location, language, or publication date. Research protocols were included if they were linked to a subsequent result publication.

Exclusion criteria included qualitative papers, discussion papers, letters to the editor, and all types of reviews. Review articles were excluded to ensure inclusion of primary quantitative studies only and to avoid overlap with primary studies identified through the literature search. Conference proceedings that did not report study findings and research protocols without a corresponding results publication were excluded, as were studies conducted in primary care settings other than CHCs. Papers focusing on the practice of a single provider (e.g., physician-led care) or a dyadic team (e.g., physician-nurse) were excluded, as well as papers examining specific interventions or programs (e.g., addition of weekly meetings for staff or smoking cessation programs for patients) within CHCs. Papers involving the pilot testing or implementation of specific new interventions that were not related to teams' day-to-day and routine characteristics, structures, models of care, or practices were excluded. Papers that measured organizational outcomes (e.g., cost) or interprofessional team outcomes (e.g., clinician communication) without assessing patient-level outcomes were also excluded.

Study Selection

A standardized chart for paper selection was developed for the assessors. The

selection process was carried out by two independent assessors (LAA, PGH). A training session was conducted using 25 titles to review the inclusion/exclusion criteria and familiarize assessors with the screening platform. Inter-rater agreement was calculated at each stage of the study selection process. Interpretation of agreement percentages (0–100%) was guided by prior work by Belur et al. (2021) and Park and Kim (2015), with thresholds defined as follows: 0–20% slight agreement, 21–40% fair, 41–60% moderate, 61–80% substantial, and 81–100% near-perfect agreement.

A calibration session with 300 titles resulted in an inter-rater agreement of 76.28% (substantial agreement)(Belur et al., 2021; Park & Kim, 2015). Titles and abstracts were screened, and full texts were retrieved if abstracts were unavailable. Duplicates were removed using the Bramer et al. (2017) technique (Bramer & Bain, 2017). The inter-rater agreement for title and abstract screening was 79.30% (substantial agreement)(Belur et al., 2021; Park & Kim, 2015). The primary area of disagreement centered on determining whether or not a study focused on TBMC characteristics. Full-text screening was also carried out by the same assessors (LAA, PGH), resulting in an 87.88% inter-rater agreement (near perfect agreement)(Belur et al., 2021; Park & Kim, 2015). Final decisions about paper inclusion were made through consensus, with a third assessor (WPW) resolving any remaining discrepancies. Ongoing meetings and discussions between the two assessors (LAA, PGH) and a tie-breaker (WPW) helped strengthen consensus and resolve divergences. A PRISMA flowchart was

developed to document the study selection process and record reasons for exclusion (Figure 1).

Data Extraction

A standardized extraction form was created and pilot-tested by the research team before use. The following data were extracted from each included paper: 1) title, 2) first author, 3) year of publication, 4) location, 5) study design, 6) research objective/question, 7) CHC description, 8) population/sample size, 9) team characteristics, 10) team composition, 11) comparison, 12) data sources, 13) tools used, 14) data collection timeline, 15) outcome measures, 16) participant-level covariates, 17) team-level covariates, 18) organizational-level covariates, 19) additional qualitative data collection, and 20) key findings. Data from secondary analyses and papers related to the same research project were extracted alongside the primary study and color coded in the extraction chart. Extraction was performed by one reviewer (LAA) and reviewed for accuracy by a second reviewer (ICA).

Data Analysis

Following the Cochrane methodological guidance for descriptive analysis in systematic reviews, descriptive statistics were used to summarize the characteristics of the included papers (Davey et al., 2011). In line with the recommendations of Popay et al. (2006), a narrative synthesis of the available evidence was conducted, and a thematic analysis was used to group findings based on TBMC characteristics (e.g., provider types, staffing patterns, electronic health record utilization, etc.) (Popay et al., 2006). Themes were identified by the

research team through an inductive process based on findings reported from the included papers. A p-value of 0.05 was set for identifying statistically significant associations (Davey et al., 2011). Meta-analysis was not attempted due to the methodological heterogeneity of the included papers (Higgins et al., 2003).

Study Quality Appraisal

The Joanna Briggs Institute (JBI) Checklists were used to assess the methodological quality of the included papers (Barker et al., 2023; Munn et al., 2023). These checklists were selected because they accommodate various quantitative study designs. The JBI checklists for cohort, cross-sectional, economic evaluation, quasi-experimental, and RCT studies were applied, with the appropriate checklist selected based on each study's design. For mixed-methods studies, the quantitative component was evaluated using the appropriate JBI checklist, as JBI does not provide a checklist specifically for mixed-methods designs. An overall score was assigned to each paper, calculated as the percentage of "yes" responses relative to the total possible points. The risk of bias and the assignment of each score were assessed by two independent reviewers (LAA, ICA). A calibration session was conducted with five full-text papers, yielding an inter-rater agreement of 92% (near-perfect agreement) (Belur et al., 2021; Park & Kim, 2015). Divergences were discussed between the reviewers to reach consensus.

RESULTS

Out of the 6,309 records initially retrieved, 5,418 were excluded after reviewing their titles and abstracts, as outlined in Figure 1. A total of 581 papers were

further screened during the full-text review, of which 548 were excluded, resulting in 33 papers being included. An additional three papers identified through other sources were also added (n=36). The primary reasons for exclusion were: 1) the study was not conducted in a CHC, and 2) it did not focus on the characteristics of TBMC. Specifically, many of the excluded papers explored the implementation of interventions involving interprofessional teams, such as introducing weekly team meetings or creating online discussion forums.

The final selection comprised 36 papers, which included 27 distinct research studies. Two papers were related to the same overarching study, another two were linked to a second overarching study, and five papers were published from a third study. The characteristics of all included papers are summarized in Table 2 and Appendix B. These papers were published between 1994 and 2022, with 19 (52.7%) published between 2010 and 2019, and 12 (33.3%) between 2020 and 2024. Most of the papers were conducted in the United States (n=27) and were based on a cross-sectional design (n=15). One research protocol was included because it provided additional methodological information not available in the results paper.

Team-based Models of Care Characteristics and Composition

TBMC characteristics were grouped into five main themes: 1) types of providers, 2) specialized interprofessional teams, 3) staffing patterns and team dynamics, 4) accreditations, and 5) e-health. The following counts are not mutually exclusive, as some papers were categorized under more than one main theme. A total of 18 papers focused on adding specific types of providers to teams. Five papers

explored specialized interprofessional teams, including social service liaison teams (n=1), weight clinics (n=1), diabetes-specialized teams (n=1), pain teams (n=1), and hepatitis C teams (n=1). Seven papers examined staffing patterns and team dynamics. Three papers examined accreditations: two focused on CHCs with Patient-Centered Medical Home Model (PCMH) certification, and one on CHCs with National Health Service Corps (NHSC) certification. Six papers examined e-health, including four investigating telemedicine utilization and two exploring the use of EHRs within CHCs.

Each included paper described one team. On average, team compositions included 5 to 6 types of providers, ranging from a minimum of 3 (inclusion criteria) to a maximum of 16 different provider types per team. Based on our inductive process and the types of providers reported in the included papers, most teams included primary care providers, which were mostly physicians (n=33), nurse practitioners (NPs), and physician assistants (PAs) (n=21), and nurses (n=20). Other providers included mental health/substance abuse/behavioral health specialists (n=12), dietitians (n=11), social workers (n=8), non-clinical/administrative/enabling staff (n=8), pharmacists (n=7), chiropractors (n=7), occupational therapist and physiotherapist (n=6), case managers or coordinators (n=5), massage therapist (n=5), medical assistants (n=4), psychologists (n=3), health promoters (n=3), vision and dental care (n=3), psychologist (n=2), midwives (n=2), interpreter (n=1), and laboratory technicians (n=1), and community health worker (n=1).

Patient Outcomes

A total of six themes of patient outcomes were identified: (1) population and disease management, (2) number of visits, (3) provider outcomes, (4) experience of care and satisfaction, (5) medication intake, and (6) pain and disability. The following counts are not mutually exclusive, as some papers examined more than one type of outcome. Fifteen papers examined population and disease management, 15 examined number of visits, six examined experience of care and satisfaction, four examined provider outcomes, four examined pain and disability, and two examined medication intake.

Sample Sizes

Average sample sizes were calculated for papers with the same unit of analysis. Among 16 papers that included patient samples, the average sample size was 267.14 (SD=310.24), ranging from 30 to 945 participants. For the 11 papers that reported the sample size of CHCs, the average sample size was 904.82 (SD=389.51), ranging from 16 to 1,384 CHCs. Four papers included the total number of patient encounters with an average sample size of 7,848.33 encounters (SD=7,796.90). Two papers examined the number of annual patient encounters per CHCs, with sample sizes of 3,139 and 4,096 encounters per year per CHCs. One paper focused on audit calls and had a sample size of 21,302 calls. Most of the included papers ($n = 22$) used electronic health data for their data collection, and half ($n = 18$) employed surveys. A total of 23 papers included a comparison group, while 13 did not. Types of comparison groups included pre- and post-measures ($n = 7$), usual care ($n = 4$), different TBMC models of care

compared to each other (n=2), different types of providers compared to each other (n=2), different measures compared to each other (n=1), funding status (n=1), medication intake by patient (n=1), private physician office compared to CHC (n=1), teaching status (n=1), PCMH (patient-centered medical home) status (n=1), case-control (n=1) and telemedicine usage (n=1).

Risk of Bias

The risk of bias was assessed for 35 included papers. No risk of bias assessment was performed on the included study protocol. On average, the risk of bias score was 65.30% (SD: 24.52), with scores ranging from 24.51% to 100%. Two papers (Garner et al., 2007; Saloner et al., 2018) achieved a top score of 100%, while four papers (Katz et al., 1998; McCarthy et al., 2021; Shin et al., 2014; Young, 1994) scored 25% or lower. A total of 18 papers (Chen et al., 2021; Chuang et al., 2019; Emary et al., 2017; Frogner et al., 2017; Garner et al., 2007; Han & Ku, 2019; Ku et al., 2015; Larrison et al., 2010; Miharti et al., 2021; Morgan et al., 2015; Morgan et al., 2014; Park et al., 2020; Park et al., 2018; Saloner et al., 2018; Shin et al., 2014; Sims et al., 2018; Vargas Bustamante et al., 2017; Young, 1994) were assessed using the JBI cross-sectional checklist, including the fifteen cross-sectional studies and three mixed-method studies in which the quantitative analysis were based on cross-section design (average score = 70.13%, SD = 23.14), 11 (Emary et al., 2022a, 2022b; Fitzgerald et al., 2021; Gold et al., 2017; Han et al., 2019; Katz et al., 1998; Luo et al., 2021; McCarthy et al., 2021; Prater et al., 2020; Shin et al., 2014; Si et al., 2006) using the cohort checklist (average score = 57.01%, SD = 29.05), three (Emary et al.,

2019; Pyne et al., 2015; Thorsen et al., 2020) using the economic evaluation checklist (average score = 72.72%, SD = 18.18), two (Fortney et al., 2013; Rodriguez et al., 2018) using the randomized controlled trial checklist (average score = 65.30%, SD = 24.51), and one (Nembhard et al., 2020) using the quasi-experimental checklist (score = 77.7%).

The following section reports the overall scores as well as strengths and weakness of papers per study design from the JBI checklists. The JBI checklists do not specify threshold scores for quality ranking; therefore, it is not possible to assign a quality category (for example, low, moderate, high risk of bias) based on the overall scores (Barker et al., 2023; Munn et al., 2023). Overall, most papers identified, accurately measured, and considered confounding variables in their statistical analyses. More than half of the papers assessed with the cross-sectional JBI checklist scored 75% or higher. However, several papers lacked detailed descriptions of the study subjects and settings. Papers assessed with the cohort studies checklist showed a wide range of scores, with half scoring above 80% and several scoring below 50%. Most of those papers did not provide information on loss to follow-up or strategies to address incomplete follow-up. All three economic evaluation papers scored higher than 50%. RCTs and quasi-experimental papers showed mixed risk of bias scores. Two papers scored above 60%, while one had a score below 40%. RCTs were particularly associated with bias risks due to the absence of concealed allocation of participants to treatment and control groups, lack of blinding to group assignment, and insufficient information on attrition and loss to follow-up.

Appendix C provides the detailed risk of bias assessment, checklists, and scores for all 35 included papers.

Summary of Findings

Through thematic analysis, five main groups of TBMC characteristics were identified: 1) type of providers, 2) staffing patterns and team dynamics, 3) specialized interprofessional teams, 4) accreditations, and 5) e-health. These categories are further elaborated in the following sections, with Table 3 summarizing all categories and outcomes. Patient outcome themes are presented under each of those five categories. Findings with statistical significance are presented along with their corresponding p-values. Results that were not subject to statistical validation are presented as reported.

Type of providers

A total of 18 papers examined the addition of specific types of providers within interprofessional teams and their association with a wide range of patient outcomes. Seven papers examined the inclusion of a chiropractor in interprofessional teams. Five papers were generated from the same research study, resulting in chiropractors being studied in three distinct settings. Garner et al. (2007) and Prater et al. (2020) identified significant associations between the addition of a chiropractor and reductions in pain and disability ($p > 0.01$), as well as improvements in participants' overall health ($p < 0.05$). Emary et al. (2017, 2019, 2021, 2022, 2022) published five papers on the same overarching research study and found significant associations between the inclusion of a chiropractor and decreases in opioid prescriptions and refills (hazard ratio: 0.48,

99% CI: 0.29–0.77), as well as reductions in pain, discomfort, and disability, and improvements in patient satisfaction and mobility (Emary et al., 2019; Emary et al., 2017; Emary et al., 2022a, 2022b; Emary et al., 2021).

Four papers examined the inclusion of NPs and PAs, also known as advanced practice providers, in interprofessional teams, which is a small number considering that these providers were included in 21 teams and the findings were mixed. In these papers, NPs and PAs managed their own patient panels under autonomous (NPs), collaborative (NPs and PAs), or supervisory (PAs) models of care involving physicians. In all papers, NPs and PAs practiced within a team-based model of care, aligned with this review's inclusion criteria, and worked in partnership with other healthcare providers. Gold et al. found a significantly higher likelihood of angiotensin converting enzyme inhibitor (ACEI) and angiotensin receptor blockers (ARB) prescriptions for patients with diabetes when managed by NPs or PAs, compared to those followed by physicians ($p<0.05$) (Gold et al., 2017). Morgan, Park, and Saloner also explored the impact of NPs and PAs on visit volume within interprofessional teams. Saloner suggested that CHCs with NPs, PAs, and physicians had a higher number of employed primary care providers compared to private offices, resulting in greater appointment availability ($p<0.01$) and shorter wait times until appointment ($p<0.01$) for uninsured patients (Saloner et al., 2018). Morgan suggested that physicians see a higher volume of patients per day compared to NPs and PAs ($p<0.05$) (Morgan et al., 2015; Morgan et al., 2014). Park examined the scope of practice for NPs and PAs and found that the number of visits per NP or PA

increased with greater prescription authority and more support staff ($p < 0.05$).

Also, the number of visits per full-time equivalent (FTE) physician decreased as NP prescription authority increased. Finally, in states with broader PA scopes of practice, there was a significant decrease in NP visits (Park et al., 2020).

The remaining papers examined the addition of a wide range of provider types within interprofessional teams, including aboriginal health workers, medical assistants, community health workers, behavioral health staff, nurse coordinators, physician residents, interpreters, and pharmacists. Aboriginal health workers, medical assistants, and community health workers were associated with improved delivery of diabetes care and management, as well as better control of low-density lipoprotein cholesterol (LDL-C) levels ($p < 0.001$) (Rodriguez et al., 2018; Si et al., 2006). Behavioral health staff were associated with an increase in visits for patients with depression and anxiety, particularly those living in rural areas (Han & Ku, 2019). For high-users and patients with severe chronic conditions, the addition of a nurse coordinator was associated with increased visit frequency and better continuity of care ($p < 0.001$) (Nembhard et al., 2020). The inclusion of physician residents within interprofessional teams was significantly associated with lower rates of low birth weight in newborns ($p < 0.01$) (Chen et al., 2021), while the addition of a Spanish interpreter was linked to higher patient satisfaction and a better care experience for patients living in the U.S. and coming from South America or Mexico (Larrison et al., 2010).

Staffing patterns and team dynamics

A total of six papers examined staffing patterns. Staffing patterns were defined by the mix and/or number of full-time equivalent (FTE) employees across various categories of providers. The most frequently reported provider categories in the included papers were primary care providers, advanced practice nurses, allied health professionals, and behavioral and mental health providers. Most of the papers identified that an increase in FTE employees was associated with a higher number of visits (Han & Ku, 2019; Ku et al., 2015; Luo et al., 2021; Thorsen et al., 2020). Luo et al. (2021) and Ku et al. (2015) suggested that higher proportions of FTE physicians within interprofessional teams were associated with increased productivity and a higher adjusted volume of patients seen ($p < 0.001$). Thorsen et al. (2020) and Han et al. (2019) found that higher proportions of FTE behavioral health staff within the entire healthcare team were linked to better management of patients with depression, anxiety, and diabetes. Chuang et al. (2019) found that lower provider-to-patient ratios were associated with higher cancer screening rates ($p < 0.05$). Miharti et al. (2021) suggested that interprofessional teams with a lower staff mix (e.g., variety of different types of healthcare providers within a team) outperformed those with a higher staff mix, being associated with increased care delivery to mothers and infants (e.g., vaccination rates, contraception use) ($p < 0.05$). One paper examined the relationship between staff workload, staff relationships, and the management of diabetes. Vargas Bustamante et al. (2017) found that a high workload alone was associated with a lower likelihood of having optimal LDL-C levels, while a high

workload combined with good staff relationships was associated with higher volumes of LDL-C, blood pressure (BP), and hemoglobin A1C (HbA1C) testing ($p<0.05$).

Specialized interprofessional teams

A total of five papers explored specialized interprofessional teams (i.e., teams that are part of standard practice and fully integrated into daily care within the CHC) focused on social liaison service inclusion (Young, 1994), weight management for children with obesity (Anand et al., 2010), diabetes (Katz et al., 1998), pain management (Prater et al., 2020), and hepatitis C (Sims et al., 2018). Young et al. (1994) found that participants followed by a specialized social liaison team reported feeling a greater sense of responsibility and engagement in their care ($p<0.05$). Anand et al. (2010) investigated a weight management clinic for children with obesity and found that, out of 402 participants, nearly 80% reported making changes to their lifestyle, physical activity, or diet, and about half showed a decrease in body mass index (BMI) after a 12-weeks follow-up. Katz et al. (1998) reported that, after intervention by a diabetes-specialized team, 66% of participants were managing their glucose at home ($p<0.001$). Additionally, most participants who had experienced hypoglycemic episodes at home saw resolution post-intervention. Among patients without hypoglycemia, mean HbA1C levels were significantly better controlled after the intervention compared to baseline measures ($p=0.02$). Prater et al. (2020) compared a specialized pain team to chiropractor care and found reductions in reported pain and disability in both groups. Participants receiving chiropractic care reported lower pain levels

compared to those seen by the pain clinic ($p < 0.01$). Sims et al. (2018) evaluated a hepatitis-C specialized team and found that 85% of patients under the team's care completed therapy (with no missed appointments), and 85% achieved a sustained virologic response.

Accreditations

Two papers in the US compared CHCs with and without the Patient-Centered Medical Home Model (PCMH) accreditation (Chuang et al., 2019; Park et al., 2018) and one paper the National Health Service Corps (NHSC) (Han et al., 2019) accreditation. The PCMH is an accreditation available for healthcare centers in the US that focuses on the delivery of team-based, patient-centered care, continuity of care, care collaboration and transitions, and quality improvement. To be certified, healthcare centers must meet all 40 core criteria (e.g., after-hours access to appointments, involving patients in treatment discussions) and at least 25 of the elective criteria. Park et al. (2018) found that in PCMH-CHCs, an increase of one FTE advanced practice staff (e.g., advanced practice nurse, NP, PA) was associated with an approximate incremental increase of 539 visits annually ($p = 0.03$) compared to CHC without the PCMH accreditation. No other significant associations were identified for any other types of providers. Elsewhere, Chuang et al. (2019) did not find any significant association between PCMH accreditation and cancer screening rates.

Han et al. (2019) compared CHC participating in the NHSC program with CHC not enrolled in program and examined whether NHSC accreditation was associated with higher number of visits. The NHSC serves as a hiring program

for CHCs, promoting the hiring of recent graduates into CHCs. Han et al. (2019) did not find any significant association between NHSC-affiliated CHCs and non-NHSC CHCs regarding the total number of visits for all types of providers.

E-health

A total of six papers examined e-health including the utilization of EHR and telemedicine in CHCs. A total of four papers examined the practice of interprofessional teams via telemedicine. McCarthy et al. (2021) explored the implementation of a telepharmacy sub-team within an on-site interprofessional team. Shin et al. (2014), Pyne et al. (2015), and Fortney et al. (2013) examined the delivery of care by interprofessional teams via telemedicine, including both on-site and off-site providers. Compared to on-site care provided by interprofessional teams alone, all papers identified significant associations between the addition of telemedicine care and improvements in several outcomes: higher monthly mean number of patient seen (McCarthy et al., 2021), better control of HbA1C for patients with diabetes (Shin et al., 2014), a greater reduction in depression symptoms, an increase in depression-free days, and overall improved wellbeing for patients with depression ($p < 0.001$; $p < 0.009$) (Fortney et al., 2013; Pyne et al., 2015).

Two papers examined the utilization of electronic health records (HER) by interprofessional teams. Frogner et al. (2017) categorized interprofessional teams by the number of years since they adopted EHR within their practice (ranging from no live EHR to seven or more years) and found that CHCs where EHR had been adopted 3 or 4 years earlier were associated with the highest

physician productivity. Chuang et al. (2019) measured EHR utilization in three subcategories: absence of EHR, basic use (e.g., capturing and sharing clinical data), and advanced use (advanced clinical processes and benefits perceived by healthcare providers) and found that EHR use was associated with higher cervical and colorectal cancer screening rates ($p < 0.01$). Forgnier et al., (2017) suggested that a 10% increase in physicians' FTE was associated with double the increase in visits in CHCs with EHR compared to those without.

DISCUSSION

The aim of this systematic review was to examine the association between TBMC characteristics and patient outcomes in CHCs. A total of 36 papers were retrieved, focusing on five themes of interprofessional teams' characteristics and six themes of patient outcomes. Most of the papers identified benefits for patients and families resulting from specific TBMC characteristics.

The findings from this review emphasize the significance of provider types within interprofessional teams and their impact on care delivery and patient outcomes. Some papers indicate that physicians are the most productive providers, with the highest volume of visits per day and per year, underscoring their key role within interprofessional teams. In recent years, the literature has identified several factors that contribute to the high productivity and patient volume of primary care physicians, including their skills and experience, the types of patients they see, and the regulations in their workplace. For instance, appointment durations may vary depending on the patient's condition and needs. Physicians also often treat both recurring patients and individuals with low-

complexity needs, which leads to shorter appointments and higher visit volumes (Kajaria-Montag et al., 2024; Morgan et al., 2014). Additionally, factors such as time pressure (i.e., the need for physicians to see patients in less time) and compensation may further influence their high volume and productivity (Freedman et al., 2021; Lee et al., 2021).

This review also highlights the value of a diverse range of providers and specialized interprofessional teams in delivering high-quality, targeted care to patients and families managing complex clinical conditions. Previous studies have emphasized the significance of aligning the scope of practice of specific providers with the unique needs of patients based on their conditions. Patient conditions often require specialized care, and each provider type brings specific skills, training, and expertise to address these needs (Khan et al., 2022). For instance, several studies have highlighted the benefits of chiropractic treatment for patients with pain, disability, and musculoskeletal disorders, which can be attributed to chiropractors' targeted approach to these issues (Trager et al., 2024). Specialized interprofessional teams were associated with improved clinical indicators for patient with diabetes, hepatitis C and childhood obesity, as the approach, provided by a diverse set of providers, addresses the multifaceted needs of patients. These teams highlight the value of a variety of provider types having specialized services, each contributing complementary expertise to care for patients with complex conditions (Bouton et al., 2023).

Some of the retrieved papers focused on telemedicine and EHR utilization, highlighting the role of electronic platforms in improving care provided by

interprofessional teams. The telehealth model of care expanded significantly during the COVID-19 pandemic and has remained widespread due to its convenience. Previous studies indicate that telehealth is particularly effective for appointments where in-person interactions are not necessary. It helps eliminate barriers such as travel and geographic distance, making healthcare appointments more accessible and convenient for patients. This increased accessibility allows for easier meetings between patients and clinicians, leading to stronger continuity of care, more follow-up appointments, and ultimately better management of patient conditions, with a reduction in negative outcomes and deterioration (Ma et al., 2022; Snoswell et al., 2022). EHRs have been in use by clinicians for decades, even before the pandemic. Research suggests that healthcare centers with EHR systems benefit from features like appointment reminders, follow-up coordination with specialists, improved communication between providers, and enhanced administrative support. These functions contribute to better clinician workflows, higher-quality care, more effective follow-up, and improved continuity of care (Nguyen et al., 2022; Rotenstein et al., 2023).

Finally, the association between accreditations and patient outcomes showed mixed results. Two out of three papers focusing on the PCMH and NHSC found no significant impact of these accreditations on patient outcomes. These findings emphasize the importance of bridging the gap between policies and actual practice in healthcare settings. Previous research has highlighted that for policies to drive meaningful change, they must have a direct impact on

bedside practice (Cantiello, 2022). To explain the lack of clear findings, other studies have suggested that certain policies and organizational structures, such as accreditation, are often reflected in other team characteristics like provider-to-patient ratios and team size. As a result, the overarching accreditation itself may not be a strong enough variable to show a direct link to patient outcomes (Han et al., 2019; Park et al., 2018).

The findings presented here should be interpreted with caution for four important reasons. First, several papers presented descriptive data, comparative groups, or pre- and post- findings without testing whether these changes were statistically significant. The absence of statistical validation raises concerns about the robustness of the findings. Second, five of the retrieved papers were linked to the same overarching research project, which could also have led to an overestimation of the findings and conclusions drawn by the authors, as well as our summary of the results. Third, in teams where certain provider types (e.g., interpreter, community health worker) or categories (e.g., behavioral and mental health provider) might be represented by small FTE proportions, relatively small additions to FTE capacity may be enough to produce measurable effects on outcomes. Finally, limited information was reported in the included studies regarding the context of the CHCs, as well as the regulatory frameworks and scopes of practice of the healthcare providers within their teams. For instance, variations in provider roles and authority (e.g., NP, PAs) across urban versus rural CHCs and teaching versus non-teaching CHCs may influence clinical practice and, consequently, patient outcomes.

Future research should focus on four key directions. First, more empirical data is needed to examine the types of providers within interprofessional teams and their association with patient profiles, including clinical, mental health, socioeconomic, and equity-related factors. The findings from this review highlight the impact of provider types on patient outcomes, but most of the included papers have focused on the general population or patients with a narrow set of conditions (e.g., diabetes). Less attention has been given to subpopulations with shared characteristics, such as individuals with low income, multimorbidity, or psychiatric disorders. More research is needed to identify the most effective provider types for these subgroups, as well as how specific providers can address the unique needs of each subgroup.

Second, while this review identified several papers that focused either on a specific type of provider or on a mix of providers, less attention has been paid to understanding the value of varied healthcare provider types, such as specialized professions, and which combinations of providers are most effective for managing patients based on their conditions. For instance, specialized interprofessional teams are designed to collaborate on treating patients with specific conditions, but models like TBMC involve a broader range of providers, and patients may see different types depending on their needs. It remains unclear which combinations of providers are most appropriate for patients with particular conditions.

Third, future research should examine outcomes related to healthcare resource utilization. Surprisingly, the papers included in this review did not examine outcomes like emergency room (ER) visits, hospital admissions, or primary care visits outside of CHCs. This may be due to the focus of the included studies on team characteristics at the individual level, assessing direct benefits for patients, clinicians, and CHCs. However, future research exploring healthcare resource utilization could provide deeper insights into how TBMC models affect access to care and whether the care provided within CHCs is sufficient. Previous research has suggested that metrics like ER visits and hospital admissions could serve as proxies for the quality and continuity of care delivered by interprofessional teams, if measured accurately (Audet et al., 2024; McDonald et al., 2024).

Finally, future research should explore a broader range of emotional and psychosocial indicators. While several studies have examined various clinical and health promotion outcomes, and some have assessed PROMs related to satisfaction and care experience, few have investigated the long-term emotional, mental, and psychosocial outcomes associated with TBMC in CHCs. For example, future studies should consider outcomes such as distress, perceived readiness and satisfaction in self-managing care, feelings of isolation and burden, and other psychosocial factors experienced by individuals living with chronic conditions or affected by social determinants of health and inequities. Investigating these outcomes would enhance our understanding of how TBMC influence the well-being of patients, caregivers, and families.

This review has several key strengths, as well as some limitations. First, to the best of our knowledge, this is the first systematic review to comprehensively examine a broad range of TBMC characteristics and patient outcomes specifically within CHC settings. In our work, we captured a variety of TBMC characteristics, including provider types, staffing, EHR and telemedicine usage, and health center accreditations. Additionally, we retrieved a wide array of outcomes, covering clinical indicators, lifestyle promotion, access to care, patient experience, and mental health indicators. Second, for this review, we conducted an extensive search across multiple sources (e.g., databases, grey literature, and secondary searches), with no restrictions on language, location, or publication year. Furthermore, we included a broad spectrum of quantitative designs and types of CHCs and similar organizations. This thorough search enabled us to gather studies from diverse geographical areas, spanning different time periods, and employing various research methodologies. However, some limitations should be acknowledged. Despite our extensive search strategy across several databases and grey literature, there is still a possibility that some relevant studies were missed. Additionally, the high heterogeneity of the results prevented us from conducting a formal meta-analysis. Finally, the aim of this review was to retrieve objective TMC characteristics; thus, we did not conduct an analysis related to team effectiveness or team functioning.

CONCLUSION

CHCs cater to a wide range of populations with complex and multifaceted needs, underscoring the significance and relevance of TBMC in these settings. This

review emphasizes the importance of TBMC characteristics and their influence on patient experience and satisfaction, the management of both clinical and psychosocial conditions, and access to care. However, to accurately assess these impacts and strengthen the evidence base, studies must employ robust methodological designs and rigorous analyses.

DECLARATIONS

Ethics approval and consent to participate

Not applicable

Consent for publication

Not applicable

Availability of data and materials

All data supporting the findings of this study are available within the paper and its Supplementary Information

Competing interests

None to declare.

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Authors' contributions

All individuals listed as authors, including L.A.A., P.G.H., I.C.A., J.R., R.M.M., and W.P.W., meet all criteria for authorship on this manuscript. Conceptualization was carried out by L.A.A., P.G.H., I.C.A., J.R., R.M.M., and W.P.W. Methodology was developed by L.A.A., P.G.H., I.C.A., and W.P.W. Visualization was performed by L.A.A. and I.C.A. Funding acquisition was led by W.P.W. Formal analysis and data curation were conducted by L.A.A., P.G.H., I.C.A., and W.P.W. The original draft was written by L.A.A., and all authors (L.A.A., P.G.H., I.C.A., J.R., R.M.M., and W.P.W.) contributed to reviewing and editing the manuscript.

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List of abbreviations

ACEI: Angiotensin converting enzyme inhibitor

ARB: Angiotensin receptor blocker

BMI: Body mass index

BP: Blood pressure

CG-CAPS1: Clinician and Group Consumer Assessment of Healthcare Providers and Systems

CHCs: Community Health Centres

CLSC: Centre local de services communautaires

CSQ-8: Client Satisfaction Questionnaire

EHR: Electronic health records

EQ-5D: EuroQoL

ER: Emergency room

FQHC: Federally qualified health center

FTE: Full-time equivalent

HbA1C: Hemoglobin A1C

HCP: Healthcare professionals

HRSA: Health Resources and Services Administration

JB: Joanna Briggs Institute

LDL-C: Low-density lipoprotein cholesterol

NP: Nurse practitioner

NHSC: National Health Service Corps

PA: Physician assistant

PCMH: Patient-centered medical home

PROMs: Patient-reported outcome measures

RCT: Randomized controlled trials

SF-12: 12-item Short Form Survey

SRQ-20: Self-Reported Questionnaire

TBMC: Team-based model of care

UDS: Uniform Data System

US: United States

WES-1: Work Environment Survey

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Figure 1. PRISMA 2020 flow diagram for new systematic reviews which included searches of databases and registers only

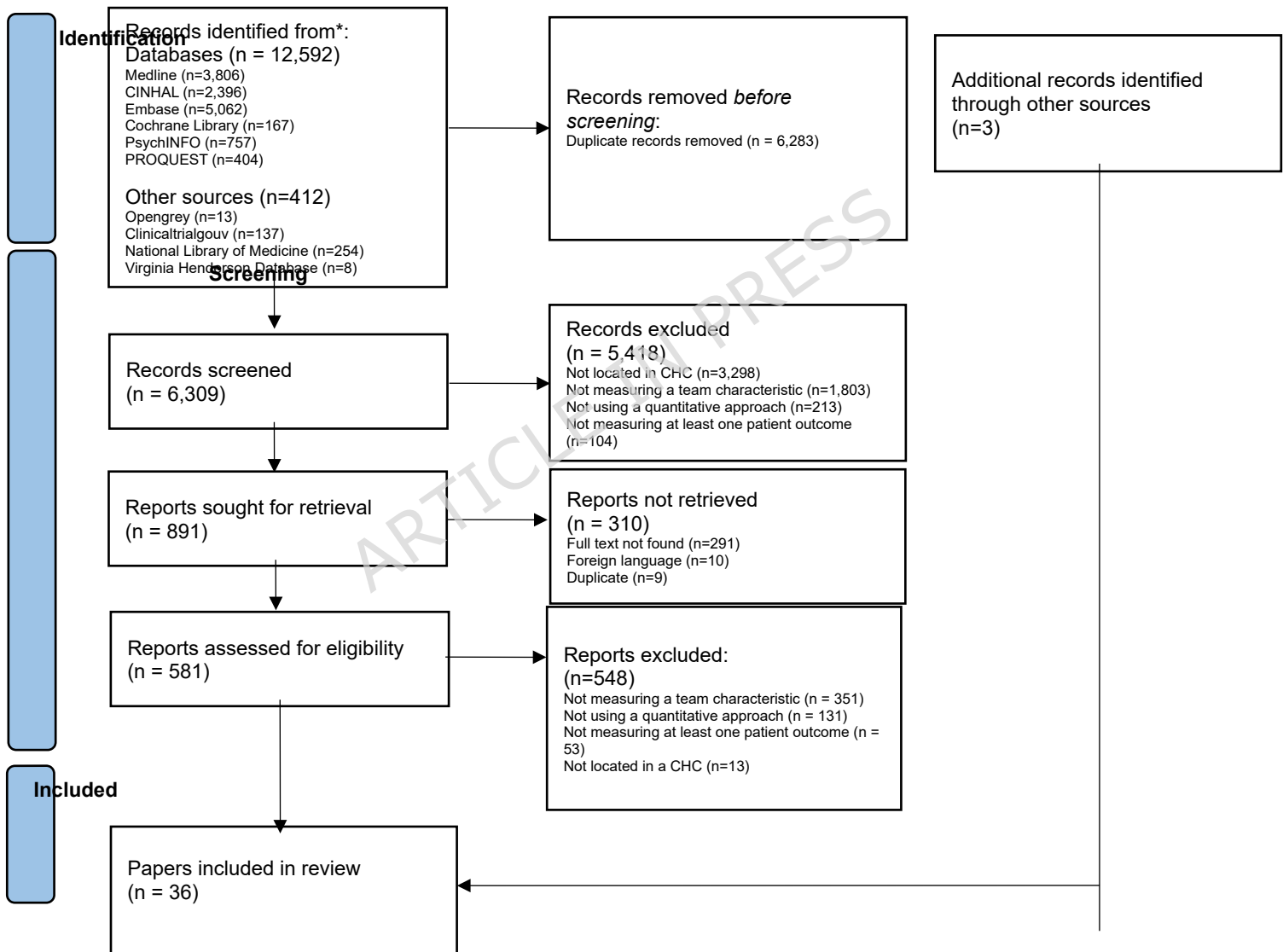


Table 1. Keywords used for the electronic search

Community health centres related keywords	TBMC related keywords
Community health center	Interprofessional team
Community health service	Collaborative care
Local community health center	Multidisciplinary team
Health and social service center	Interdisciplinary team
Federally qualified health center	Clinical team
CHC, FQHC, CLSC	Primary health care team
	Cooperative team

Table 2. Summary of Descriptive Results

Variable	Number of papers
Country of publication	
<i>US</i>	27
<i>Canada</i>	6
<i>UK</i>	1
<i>Australia</i>	1
<i>Indonesia</i>	1
Design	
<i>Cross-sectional</i>	15
<i>Cohort</i>	8
<i>Mixed-method</i>	4
<i>RCT</i>	3
<i>Pilot</i>	2
<i>Quasi-experimental</i>	1
<i>Economic evaluation</i>	1
<i>Service evaluation</i>	1
<i>Research protocol</i>	1
Unit of analysis	
<i>Individual participant</i>	14
<i>CHC</i>	11
<i>Providers + participants</i>	4
<i>Patient encounters</i>	2
<i>Patient + provider encounters¹</i>	2
<i>CHC annual observation²</i>	2
<i>Audit calls</i>	1
Population	16
<i>General population</i>	5

<i>Back and neck pain</i>	2
<i>Depression</i>	2
<i>Diabetes</i>	2
<i>Hepatitis C</i>	2
<i>Diabetes on Chinese and Latino populations</i>	1
<i>Elderly patients³</i>	1
<i>“High risk” patients⁴</i>	1
<i>South America or Mexico populations</i>	1
<i>Musculoskeletal conditions</i>	1
<i>Rural residents</i>	1
<i>Chronic spinal pain</i>	1
<i>Overweight children</i>	1
Data Collection with EHR	22
<i>Uniform Data Service and Health Resources and Services Administration (US national databases)</i>	12
<i>National database in Indonesia</i>	1
<i>EHR of their affiliated CHC</i>	9
<i>Chart review</i>	1
<i>Audit calls</i>	1
Data Collection with Surveys	18
Questionnaires developed by the research team	11
<i>Provider survey</i>	4
<i>Patient survey</i>	3
<i>Provider + patient survey</i>	1
<i>Depression-specific survey</i>	1
<i>Pain and disability scale</i>	2
Validated Questionnaires	17
<i>Patient-Centered Medical Home (PCMH) survey</i>	4
<i>EuroQoL-five dimensions Questionnaire</i>	3
<i>12-item Short Form Survey (SF-12)</i>	2
<i>Client Satisfaction Questionnaire (CSQ-8) survey</i>	1
<i>Self-Reported Questionnaire (SRQ-20)</i>	2
<i>Clinician and Group consumer Assessment of HealthCare Providers and Systems (CG-CAPS-1)</i>	1
<i>Work Environment Survey (WES-1) Roland Morris survey Bournemouth and Bothersomeness survey</i>	1
Comparison group	23
<i>Pre and post measures</i>	7
<i>Usual care</i>	4
<i>Models of care</i>	2
<i>NP/PA compared to physicians</i>	2
<i>Several measures compared to each other</i>	1
<i>Funding status (CHC fully funded vs. partly funded)</i>	1
<i>Medication intake (opioid intake vs. non-opioid intake)</i>	1

<i>Private physician office (CHC vs. private physician office)</i>	1
<i>Teaching status (yes/no)</i>	1
<i>PCMH status (yes/no)</i>	1
<i>Case-control (patient who did and didn't achieve the desired outcome)</i>	1
<i>Telemedicine usage (yes/no)</i>	1

Notes:

¹Data collected from both patients and providers for a single or several encounters

²Number of encounters per year

³75 Y.O. and older

⁴Frequent ER visits and hospitalization, multimorbidity

Abbreviations: CHC, Community Health Centre; HER, Electronic Health Record; NP, Nurse Practitioner; PA, Physician Assistant; PCMH, Patient-Centered Medical Home; RCT, Randomized Controlled Trial; UK, United Kingdom; US, United States

Table 3. Summary of papers according to TBMC characteristics and outcomes

Themes of Outcomes	Themes for TBMC exposures				
	Types of providers	Specialized interprofessional teams	Staffing pattern and team dynamics	Accreditations	e-health
Population and disease management Diabetes Expecting mothers, infants and children Depression Hepatitis C Cancer screening Lifestyle modification (physical activity and diet)	Ross D (AHW) Rodriguez H (medical assistant and CHW) Chen C (resident) Fitzgerald L (pharmacist)	Kartz G (diabetes) Sims O (hepatitis C) Shikha G (overweight children)	Thorson M (% of FTE, staff mix) Luo Q (staffing pattern, % of FTE) Vargas Bustamante A (workload, staff relationship) Miharti S (staff mix) Chang E (staffing ratio)		Shin P (general telemedicine usage) Payne J (general telemedicine usage) Fortney J (general telemedicine usage)
Productivity/number of visits	Nembhard I (nurse coordinator) Chen C (resident) Han X (behavioral health staff) Saloner B (NP, PA) Park J (NP, PA) Morgan P(a,b) (NP, PA)		Miharti S (staff mix) Ku L (staff mix, % of FTE) Han X(b) (% of FTE)	Park J(b) (PCMH) Han X(b) (NHSC) Chang E (PCMH)	Frogner B (EHR) McCarthy C (telepharmacy) Chang E (EHR)
Provider outcomes	Nembhard I (nurse coordinator) Larrison C (interpreter) Chen C (resident)	Young K (social liaison)			
Experience of care and satisfaction	Nembhard I (nurse coordinator) Larrison C (interpreter) Garner M (chiropractor) Rodriguez H (medical assistant and CHW)	Young K (social liaison)			McCarthy C (telepharmacy)
Medication intake	Emery P (a,b,c,d) (chiropractor) Gold R (NP, PA)				
Pain and disability	Garner M (chiropractor) Prater C (pain team vs chiropractor) Emery P (a,b,c,d) (chiropractor)	Prater C (pain team vs chiropractor)			

Abbreviations: AHW, aboriginal health worker, CHW, community health worker, EHR, electronic health records, FTE, full-time equivalent, NHSC, National Health Service Corps, NP, nurse practitioner, PA, physician assistant, PCMH, patient-centered medical home

Appendices

Appendix A. Electronic Database Search Strategy and Grey Literature

Database: Medline, Embase, APA Psycinfo

1. (Communit* health adj3 (center* OR centre* OR service*)).ti.
2. (Local communit* adj3 (center* OR centre* OR service*)).ti.
3. (Health and social service adj3 (center* OR centre*)).ti.
4. (Federal qualified health adj3 (center* OR centre*)).ti.
5. (CHC* or FQHC* or CLSC*).ti.
6. 1 or 2 or 3 or 4 or 5
7. (Primary adj3 (care or health care)).ti.
8. (Community adj3 (care or health care)).ti.
9. (Family adj3 (medicine or practice)).ti.
10. 7 or 8 or 9
11. (Team* adj2 (patient care OR interprofession* OR cooperat* OR collaborat* OR transprofession* OR multidisciplin* OR multiprofession* OR interdisciplin* OR clinical OR primary care OR primary health care)).ti.
12. 10 and 11
13. 6 or 12

Database: CINAHL

- S1 TI(Communit* health N3 (center* OR centre* OR service*))
- S2 TI(Local communit* N3 (center* OR centre* OR service*))
- S3 TI(Health and social service N3 (center* OR centre*))
- S4 TI(Federal qualified health N3 (center* OR centre*))
- S5 TI(CHC* or FQHC* or CLSC*)
- S6 S1 or S2 or S3 or S4 or S5
- S7 TI(Primary N3 (care or health care))
- S8 TI(Community N3 (care or health care))
- S9 TI(Family N3 (medicine or practice))
- S10 S7 or S8 or S9
- S11 TI(Team* N2 (patient care OR interprofession* OR cooperat* OR collaborat* OR transprofession* OR multidisciplin* OR multiprofession* OR interdisciplin* OR clinical OR primary care OR primary health care))

S12 S10 and S11

S13 S6 or S12

Database: PROQUEST

S1 (title((Communit* health NEAR/3 (center* OR centre* OR service*))) OR title(Local communit* NEAR/3 (center* OR centre* OR service*))) OR title((Health and social service NEAR/3 (center* OR centre*))) OR (Federal qualified health NEAR/3 (center* OR centre*) OR title(CHC* OR FQHC* OR CLSC*))

S2 title(primary NEAR/3 (care OR health))) OR title((community NEAR/3 (care OR health))) OR title((family NEAR/3 (medicine OR practice)))

S3 title((Team* NEAR/2 (interprofession* OR cooperat* OR collaborat* OR transprofession* OR multidisciplin* OR multiprofession* OR interdisciplin* OR clinical)))

S4 S2 and S3

S5 S1 and S3

S6 S4 or S5

Database: Cochrane Library

Community health centers
Community health centres

Grey literature search strategy and screening process

Source: Opengrey

Keywords : Community health centres

N=13 results

Included = 0

Source: Clinicaltrialgouv

Keyword : Community health centres

AND interprofessional team n=5 results

AND clinical team n=22 results

AND multiprofessional team = 2 results

AND primary care team n=23 results

AND patient care team n=33 results

Federally qualified health center n=43 results

AND collaborative team n=9 results

Included = 0

Source: Googlescholar

Community health center
 interprofessional teams
 multidisciplinary teams
 team characteristics
 federally qualified health center
 CHC
 FQHC
 Included = 0

Source: National Library of Medicine
 (community health cent*)
 AND (interprofessional team) n=99 results
 AND (interprofessional team composition) n=66 results
 AND (interprofessional team characteristics) n=89 results
 Included = 0

Source: Virginia Henderson Dababase
 Title(community health cent*) result = 3
 Title(federally qualified health center) result = 3
 Title(CLSC) result = 0
 Title(health and social center) result = 0
 Title(CHC) result = 1
 Title(FQHC) result = 1
 Title(community mental health cent*) result = 0
 Included = 0

Appendix B. Characteristics of the included papers

Title, first author, year of publication, location	Design and Risk of bias (%)	Study objectives	Setting	Population and sample size	Data collection (data source, tool, timeline, complementary qualitative analysis)	Independent variable - Team characteristic, composition, comparison	Comparison	Dependent variable - Outcome measured	Covariates (patient, team, organizational)	Main findings (quantitative)
Aboriginal is health workers and diabetes care in remote community health centres: a mixed method analysis <i>Ross, D. (2006)</i> <i>Australia</i>	Cohort study 90.9%	Examine the association between the employment of aboriginal health workers (AHWs) on diabetes care in remote community health centers (CHCs). Identify the barriers to the integration of AHWs in interprofessional team caring for patient with diabetes and other chronic illnesses.	2 CHCs located in remote areas of Northern Territories Australia (Tiwi Island, Katherine West Region)	Client served at the 2 CHCs of the Northern Territories with a diagnosis of type 2 diabetes Sample size = 146 participants	Data source: paper and electronic clinical records Timeline: audit of clinical records conducted at baseline, after 6 months, 1,2 and 3 years Additional qualitative data were collected to explore the facilitators and barriers of the integration of the AHWs in the interprofessional team and deepen the understanding of the quantitative findings	Addition of AHWs within interprofessional teams: o Number of AHWs per 1,000 residents o Presence of male AHWs <u>Team composition:</u> o AHWs o Physician o Nurse	Previous measures	Adherence to the delivery of guideline-schedule diabetes care (n=29): % of scheduled services provided out of 29 Most recent glycated hemoglobin (HbA1C) Blood pressure (BP) levels measured in the past 12-months before each audit	<u>Patient-level:</u> o Sex o Age o Number of comorbidities (0, 1, 2, or more) <u>Team-level:</u> o Doctor type (visiting, resident) o Average number of nurses per 1,000 residents o Number of nurses per 1,000 residents (1-4; 5 or more) <u>Organizational-level:</u> o Presence of separate men clinic o Presence of separate women clinic o % of services recorded in computerized information system (<50%; 50-70%; 70 or more) o Size of the population served (500 or less; 501-999; 1,000 or more)	Health centers with 10 or more AHWs per 1000 residents provided significantly more guideline-recommended services compared to centers with fewer than five AHWs per 1000 residents (adjusted mean difference: 17%; 95% CI: 8%–26%). The presence of male AHWs was associated with a higher delivery of guideline-based diabetes services (adjusted mean difference: 6%; 95% CI: 1%–10%). No significant association was found between the employment of AHWs and HbA1c levels or blood pressure control (p>0.05).
Adoption of telepharmacy within a community health center: A focus on clinical pharmacy service <i>McCsrthy C. (2021)</i> <i>USA</i>	Cohort study 18.8%	Describe how telehealth has been used in the delivery of pharmacy services in CHCs. Examine the effect of telehealth on the productivity of	The Henry J. Austin federally qualified health center (FQHC) located in Trenton New Jersey	Patient who had an encounter at the Henry J. Austin FQHC. The unit of analysis was the patient encounter. Patients with all types of conditions and diagnoses were included. Sample size= Pre-intervention	Data source: Reports on the number of encounters retrieved from the patient electronic health records (EHR) Tool: Patient satisfaction survey	Implementation of telepharmacy services to provide clinical care through a comprehensive medication management approach by the pharmacist and other members of the interprofessional team. The telepharmacy service was implemented in March 2020. <u>Team composition:</u> o Primary care provider o Nurse	Pre and post comparison: 7-months pre-and post-implementation	Monthly productivity defined as the monthly number of patient encounter Patient satisfaction	<u>Patient-level:</u> o Age o Sex o Language o Ethnicity o Race o Zip code o Insurance (yes/no) <u>Team and organizational-level:</u> none measured	For all types of patient encounter, overall productivity remained relatively consistent, increasing from 36,542 visits during the pre-implementation period to 36,948 visits during the post-implementation period Between the pre- and post-implementation

		interprofessional teams and patient satisfaction.		period=36,542 encounters Post-intervention period=36,948 encounters	Provider experience survey Timeline: Pre-implementation period: August 1 st , 2019, to February 29, 2020 Post-implementation period: April 1 st , 2020, to October 31 st , 2020	○ Dietician ○ Behavioral health consultants ○ Pharmacy team (outpatient pharmacy; one part-time pharmacist; two full-time clinical assistant professors; one lead clinical pharmacist; director of pharmacy)				period, there was a decline in the average monthly number of patient encounters for dental, pediatric, and podiatric services. After the implementation of the telehealth service, productivity improved for adult medicine, behavioral health, clinical pharmacy, family medicine, and nutrition services. Despite a reduction in full-time equivalent (FTE) staffing for pharmacy visits from 1.6 to 1.2, the productivity of clinical pharmacy encounters increased following the adoption of telepharmacy services. The monthly average of clinical pharmacy encounters rose from 159.3 visits during the seven months prior to implementation to 212.3 visits during the seven months afterward, representing a 33.3% relative increase. All patients reported that telepharmacy appointments were more convenient than in-person visits and that the quality of telepharmacy visits was equal to or better than in-person appointments.
An evaluative study of a community health service development <i>Young K. (1994)</i> <i>England</i>	Cross-sectional 12.5%	Compare the community health services provided by fully funded and partially funded community care groups in rural areas for individuals over 75 years old.	Fully and partially funded community care groups serving rural areas in England operated across 12 rural health districts. These districts were divided into 16 loosely defined "patches,"	Community care group worker People over 75 years old who received care from the community care team Sample size=not mentioned. The authors mentioned	Tools: <u>Provider questionnaire</u> Staff Questionnaire (Developed by the Research Team): Multi-Disciplinary	Collaboration between multi-disciplinary teams and social services liaisons in fully and partially funded community care groups. <u>Team composition:</u> ○ GP ○ District nurse Health visitor ○ Community psychiatric nurse	Funding status (fully and partly funded)	Provider Outcomes: Job satisfaction. Full-time equivalent (FTE) staff-to-population ratio. Availability of equipment and	<u>Patient-level:</u> ○ Age ○ Sex ○ Living situation (living alone) ○ Owning a home ○ Owning a car ○ Self-reported health <u>Team-level:</u> ○ Contact with health	<u>Provider Outcomes</u> Analysis of the ratios of whole-time-equivalent staff per unit population revealed comparable staffing levels in fully and partially funded community care groups. Additionally, population profiles showed no consistent disadvantage

			organized around GP caseloads. Four of these patches, located entirely within a large city, were excluded from this study. Those community care groups were serving people over 75 years old.	the response rate for the multi-disciplinary teamwork questionnaire (77%) and the social services questionnaire (63%) sent to the community care group workers. No information on the sample size of the patient who completed the patient questionnaire.	Teamwork (Developed by the Research Team) Job Satisfaction Social Services Questionnaire <u>Patient questionnaire</u> Patient Satisfaction Timeline: one-time measure in 1988 Additional qualitative data were collected with providers from both groups.	○ NHS occupational therapist ○ SSD occupational therapist ○ Physiotherapist ○ Chiropodist ○ Health promoter officer ○ Voluntary coordinator ○ Community nurse ○ Paramedic ○ Social worker ○ Home care organizer ○ Speech therapist ○ Dietician ○ Home help		material resources. Frequency of interactions between primary care teams and social services liaison teams. Experience and attitude to multidisciplinary work Patient Outcomes: Access to primary care providers and involvement in decision-making processes. Sense of autonomy and responsibility in managing their care. Engagement with providers in discussions about maintaining or improving their health.	service ○ Frequency of contact ○ Contact with different types of providers <u>Organizational-level:</u> none measured	among those served by partially funded groups. Population profiles showed no consistent disadvantage among those served by partially funded groups. Staff in partially funded groups were statistically more likely ($P < 0.05$) to feel that their work involved significant wasted effort, a component of the job satisfaction index, although this did not impact overall job satisfaction scores. Similar levels of liaison were found between social services and health services in both group types ($p > 0.05$). <u>Patient Outcomes</u> Elderly patients in partially funded groups more frequently reported difficulty implementing advice, but this difference was not statistically significant ($P < 0.05$). Elderly individuals served by fully funded groups more often believed they had greater responsibility for their health and attributed illnesses to personal neglect compared to those in partially funded groups ($P < 0.05$). Additionally, health workers in fully funded groups tended to discuss maintaining or improving health with elderly patients slightly more often than those in
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										partially funded groups, although this difference did not reach statistical significance ($P < 0.10$).
A quasi-experiment assessing the six-months effects of a nurse care coordination program on patient care experiences and clinician teamwork in community health centers <i>Nembhard I. (2020)</i> USA	Quasi-experimental 77.7%	Examine the early (six-month) impacts of a nurse care coordination program implemented in FQHC practices that adopt the added-role approach for high-risk patients. Assess these effects using two measures: the care experiences of the patients and clinician-reported teamwork.	Multi-site FQHC consisting of 12 locations, which collectively provide comprehensive primary medical, dental, and behavioral healthcare services to over 140,000 patients annually.	Adult patient (> 18 y.o.) identified as "high-risk": These patients were 18 years or older, had two or more visits with a primary care provider (PCP) in the past 12 months, and were classified as "high risk." High-risk patients met one or more of the following criteria: A) two or more emergency room visits in the past 12 months; B) one or more hospitalizations in the past 12 months; C) a Type 2 diabetes diagnosis on their problem list and a hemoglobin A1C test greater than 9% in the past 12 months; D) a diagnosis of persistent asthma on their problem list and two or more asthma control test scores below 19 in the past 12 months; or E) four or more specified chronic illnesses on their active problem list, including Type 2 diabetes, chronic obstructive pulmonary disease, hypertension, asthma, coronary artery disease, or a behavioral health diagnosis. Sample size - Baseline=3007 - Follow-up=2101 Staff: 39% PCPs,	Tool: <u>Patient questionnaire</u> The Consumer Assessment of Healthcare Providers and Systems Clinician & Group (CG-CAHPS) survey The Patient-Centered Medical Home (PCMH) Supplemental Item Set <u>Staff questionnaire</u> Organizational assessment survey Timeline: 6 months prior to the implementation of the intervention and follow-up measure 6-months after the implementation of the intervention	Addition of a nurse coordinator to the interprofessional team: In the intervention sites, the role of each nurse was expanded to include care coordination for adult patients who were most likely to benefit from this initiative. As part of the new program, nurses were tasked with assisting enrolled patients in navigating their healthcare and leading a weekly panel management meeting with the patients' PCP and behavioral health provider. <u>Team composition:</u> - PCP - Medical assistant - Behavioral health promoter - Nurse coordinator	Pre and post comparison: 6 months pre and post measures	Patient care experience Teamwork Office visit frequency	<u>Patient-level:</u> - Age - Gender - Education - Race/ethnicity - Overall health status - Mental health status - Medicare status <u>Team-level:</u> - Profession - Gender - Full-time status - Years in the organization <u>Organizational-level:</u> none measured	<u>Patient outcome:</u> No statistically significant difference in changes between the intervention and comparison groups ($p > 0.10$). Patient care experience scores increased in the intervention group (2.75 to 2.88), while scores in the comparison group remained relatively unchanged (2.82 to 2.80). The changes were not statistically significant ($P = 0.07$). Clinician-reported teamwork increased in the intervention group (3.51 to 3.60) and decreased in the comparison group (3.49 to 3.38). However, the difference was not statistically significant ($P = 0.12$). Patients enrolled in the program had 1.33 more visits than those in the comparison group, a statistically significant difference ($P < 0.001$).

				22% nurses, 24% medical assistants, and 15% behavioral health providers.						
Brokering Language and Culture: Can Ad Hoc Interpreters Fill the Language Service Gap at Community Health Centers? <i>Larrison C. (2010)</i> USA	Mixed method 62.5%	Examine the ability of the ad hoc interpreter to adapt the organizational culture of the CHCs. Assess the satisfaction of the clients who required and used the services of ad hoc interpreters with the medical care they received. Compare the practice of ad hoc interpreters to professional medical interpreters.	The CHC is in the Midwest, USA and includes regions encompassing a Metropolitan Statistical Area (MSA) along with adjacent rural areas connected by proximity and commuting patterns. The MSA housing the CHC has an estimated population of 175,000 and is surrounded by a large rural region characterized by economic challenges in agriculture and manufacturing. Within the MSA, the population is 78% White, 11.5% African American, 0.2% Native American, 6.5% Asian American, and 3.8% Latino.	Participants were originally from South or Central America or Mexico and were adults (aged 18 to 65) who had either personally received medical services at the CHCs or had an immediate family member who had done so, agreeing to participate in the study. Staff were recruited if their practice was at least quarter time at the CHC and regular contact with Latino clients. Sample size= 30 participants and 17 staff	Tool: <u>Staff questionnaire</u> Work Environment Scale-Real (WES) <u>Participant questionnaire</u> Client Measurement Instrument: Timeline: one-time measure with staff and participants Additional qualitative data collection: A focus group with the ad hoc interpreters was held in December 2005. The session lasted approximately one hour and was facilitated by two members of the research team, one of whom was bilingual and bicultural. Open-ended questions were used to explore the interpreters' perceptions of the agency.	Addition of an ad hoc interpreter into the interprofessional team: At the time of data collection, the CHC employed three interpreters. On average, they had been working with the agency for about five years. All interpreters were bilingual and bicultural, with most originating from countries outside the United States. <u>Team composition:</u> - Administrators - Direct service providers - Support staff	No comparison, single case descriptive study	Integration of ad hoc interpreter into the CHC organizational climate Client satisfaction	<u>Patient-level:</u> - Education - Sex - Age - How the client found out about the CHC services <u>Team and organizational-level:</u> none measured	Staff reported moderately higher-than-average levels of cohesion (M = 5.8), support (M = 5.5), autonomy (M = 6.1), and innovation (M = 4.7), alongside lower-than-average levels of work pressure (M = 4.2) and control (M = 4.2). The ad hoc interpreters experienced 60% (2.8 points) higher levels of work pressure than other staff and 23% (1.3 points) higher than the norm. Ad hoc interpreters reported 41% (2.1 points) less involvement with the organization, 74% (2.9 points) less clarity regarding rules and policies, 51% (3.1 points) lower task orientation, and 79% (3.7 points) fewer opportunities for innovation compared to other staff. The average Client Satisfaction Scale (CSQ-8) score was 30.23 (SD = 2.97), ranging from 22 to 32. Ten percent (n = 3) of participants scored below "satisfied with services." There were no significant differences in CSQ-8 scores based on gender or educational level. Clients unanimously reported improved

										health and family life due to the CHC's assistance. None indicated that their issues remained unchanged or worsened. When asked specifically about the ad hoc interpreters: 17% (n = 5) valued the interpreters' ability to speak their language, 3% (n = 1) valued the interpreters' knowledge of U.S. culture and 80% (n = 24) prioritized the interpreters' ability to communicate their needs effectively.
Cardiovascular care guideline implementation in community health centers in Oregon: a mixed-methods analysis of real-world barriers and challenges <i>Gold R. (2017)</i> USA	Mixed method (clustered RCT and ethnographic evaluation) 81.81%	Describe the multi-level factors influencing the implementation of guideline-based cardioprotective prescribing.	The CHCs were part of the OCHIN, Inc., a non-profit organization that provides health information technology, including the EpicCare® electronic health record (EHR). The 10 study CHCs are ambulatory primary care clinics operated by three Federally Qualified Health Center systems in the Portland, Oregon metropolitan area. These clinics vary in size and organizational structure: one is run by a large academic medical center, six by an urban county health department, and four by a non-profit organization that primarily	All in-person encounters between June 2011 and May 2014 at the intervention CHCs where an ALL-related point-of-care alert was triggered, and the provider had prescribing privileges. According to the alerts' criteria, these encounters involved adult patients (aged 18–75) with diabetes who were clinically indicated for a targeted medication based on national guidelines but had not received a prescription for it in the past year. Sample size=21,475 encounters including 11,588 statin encounters and 9,887 ACEI/ARB encounters	Data source: EHR data from the 24-months following the activation of the alerts in the study CHC Timeline: 24-months follow-up between June 2011 to May 2014 Additional qualitative data collection	Type of providers who conducted the encounter: physician, resident, NP, PA The status of the provider who conducted the encounter: primary care provider status <u>Team composition:</u> ○ Physician ○ Resident ○ NP ○ PA	Patient who had an encounter but did not receive a prescription of statin or ACEI/ARB in the last year.	Prescription of cardioprotective statins and ACEI/ARBs to patients with diabetes	<u>Patient-level:</u> ○ Sex ○ HbA1C levels ○ Prior prescription of statins and ACEI/ARB ○ Diagnosis and reason for encounter ○ Insurance status ○ Number of visits in the last year (1 to 3; 4 and >) ○ Kidney disease ○ Insulin therapy ○ Narcotic use ○ cardiovascular medication use ○ Psychotherapeutic medication use ○ Age (40 to 54; 55 to 75 y.o.) ○ LDL levels ○ Liver disease ○ Other anti-hyperglycemic medication use ○ BMI ○ Blood pressure level ○ Cardiovascular comorbidities ○ Race/ethnicity ○ Language	The likelihood of prescribing the indicated statins and/or ACEI/ARBs was significantly higher when the patient was seen by a NP or PA compared to a physician or a resident. The odds of prescribing the indicated statins and/or ACEI/ARBs were significantly higher when the patient was seen by their PCP rather than another provider. The odds of prescribing statins and ACEI/ARBs were significantly lower when the patient had one or more visits to the same clinic in the past year, compared to those who had no visits or had never been seen at the clinic before.

			serves suburban, Spanish-speaking populations.						<u>Team-level:</u> - Type of provider - Primary care provider status <u>Organizational-level:</u> none measured	
Changes in primary care access at community health centers between 2012/2013 and 2016 <i>Saloner B. (2019)</i> USA	Cross-sectional 100%	Examine how recent policy changes under the ACA have impacted the availability of primary care services at CHCs in comparison to private physician offices.	CHCs in 10 states during 2012 (pre-ACA expansion) and 2016 (post-ACA expansion): Arkansas, Georgia, Iowa, Illinois, Massachusetts, Montana, New Jersey, Oregon, Pennsylvania, and Texas. The ACA notably increased CHC grant funding by \$11 billion through the establishment of the Community Health Center Fund. This funding boost, extended through fiscal years 2016 and 2017 under the Medicare Access and CHIP Reauthorization Act of 2015 (MACRA), aimed to expand CHC capacity, allowing them to serve more patients and offer additional services such as dental care and behavioral health	Each observation of the dataset was a call to office to assess the services availability of CHCs and private physician offices. Calls were made to CHCs and private offices by members of the research team. Sample size=21,302 calls 2012/2013 CHCs=557 Private office=11,371 2016 CHCs=663 Private office=8,711	Data source: audit calls Timeline: 2012/2013 and 2016	Type of provider (physician, NP, PA) Number of general providers <u>Team composition:</u> - Physician - NP - PA	CHCs compared to private physician offices 2012/2013 compared to 2016	Appointment rate and wait time with a GP: Callers requested an appointment with a specific physician known to practice at the office but were also willing to accept an appointment with any available primary care provider, including a nurse practitioner or physician assistant. They recorded whether an appointment could be scheduled (coded as a binary variable) and, if successful, the number of days until the first available appointment	<u>Patient-level:</u> none measured <u>Team-level:</u> - Number of physicians, NP, PA - Number of non-physician clinicians (NP and PA) compared to 2.8 and 1.2, respectively, at private offices. - Mean daily patient volume - Typical volume of patient per day at the office - participation in Medicaid management care <u>Organizational-level:</u> characteristics of the offices' counties in the baseline year: - Uninsured rate - Rurality rate - Physician-to-patient ratio - Median income of the office counties - Medicare-Medicaid primary care fee ratio (created by the Urban Institute)	In 2012/2013, CHCs in the sample had more staff than non-CHC offices, with 4.2 physicians and 2.9 nonphysician clinicians (NP and PA) compared to 2.8 and 1.2, respectively, at private offices. CHCs served more patients daily, averaging 84 patients per day versus 53 at private offices. For individuals with employer-sponsored insurance (ESI), appointment availability at CHCs improved significantly from 80.0% in 2012/2013 to 92.2% in 2016 (P = 0.001), while availability at private offices remained relatively unchanged (84.2% in 2012/2013 vs. 83.4% in 2016, P = 0.543). The relative improvement at CHCs compared to private offices was statistically significant (difference-in-differences: 13.5 percentage points, P = 0.008). For Medicaid patients, appointment availability increased at CHCs from 78.0% in 2012/2013 to 88.3% in 2016 (P = 0.016) and in private

										offices from 56.3% to 61.1% during the same period ($P = 0.039$). However, the difference-in-differences was not statistically significant. For uninsured individuals willing to pay the full cost of a visit, appointment availability at CHCs remained steady (86.4% in 2012/2013 and 89.0% in 2016, $P = 0.139$) and showed a slight increase in private offices (76.9% in 2012/2013 vs. 80.4% in 2016, $P = 0.518$). The difference-in-differences was also not statistically significant. For callers with employer-sponsored insurance, the average wait time for an appointment at CHCs decreased significantly, from 15.1 days in 2012/2013 to 10.1 days in 2016 ($P = 0.048$). In private offices, the wait time increased slightly from 12.2 days to 13.0 days ($P = 0.389$). The difference-in-differences (-5.7 days) was statistically significant ($P = 0.017$), indicating a reduction in wait times at CHCs relative to private offices. For Medicaid patients, mean wait times at CHCs remained consistent (14.0 days in 2012/2013 vs. 14.4 days in 2016, $P = 0.836$), as did wait times at private offices (13.8 days vs. 14.3 days, $P = 0.601$). The difference-in-
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										differences was not statistically significant. For uninsured callers, wait times at CHCs showed no significant change (11.4 days in 2012/2013 vs. 11.6 days in 2016, $P = 0.933$), while private offices wait times increased slightly (11.2 days to 13.2 days, $P = 0.087$). The difference-in-differences was not statistically significant.
Chiropractic care of musculoskeletal disorders in a unique population within Canadian community health centers <i>Garner M. (2007)</i> <i>Canada</i>	Cross-sectional study 100%	Evaluating the effectiveness of chiropractic care in reducing pain and disability and improving general health outcomes among a unique population of urban, low-income, and multiethnic patients with musculoskeletal (MSK) complaints.	Chiropractic clinics were set up in two urban Canadian CHCs, both serving comparable populations in terms of size and sociodemographic characteristics. These neighborhoods are characterized by low socioeconomic status and diverse multiethnic communities that face significant barriers to accessing adequate health care.	All individuals who visited the clinics between August 1, 2004, and December 31, 2005, were screened for eligibility. Patients included had the inability to privately pay for chiropractic care, residence within the catchment area of either site, and the presence of a musculoskeletal (MSK) disorder. Sample size=250 participants	Tools: Pain levels were measured Roland-Morris Questionnaire (RMQ) Neck Disability Index (NDI) Short Form-12 version 2 (SF-12v2). Patient satisfaction with chiropractic care was assessed using a tool specifically developed for this study. Timeline: 2 measures before and after 12-weeks of care or at discharge, whichever came first	Addition of a chiropractor in the interprofessional team: the type of treatment and frequency of visits were determined by the treating chiropractor. The only restrictions placed on the chiropractors were the use of modalities (which were unavailable) and nutritional supplementation. Patients received treatment and were followed for a maximum of 12 weeks as part of the study, although some may have continued treatment after discharge from the study. <u>Team composition:</u> - o Physician - o NP - o Dietician - o Health promoter - o Social worker - o Counselors - o Other clinicians (no details provided)	Pre and post comparison	Pain Disability Neck pain Shoulder disability and pain Knee disability and pain Wrist disability and pain General health status Patient satisfaction	<u>Patient-level:</u> - o Sex - o Self-reported BMI - o Household income (< or > 20K) <u>Team-level:</u> None measured <u>Organizational-level:</u> - o Clinic site	Clinically significant and statistically meaningful changes were observed in both current pain (mean score = 2.3, 95% CI = 1.9-2.6) and typical pain (mean score = 2.0, 95% CI = 1.7-2.3) across all patients. Post hoc analyses revealed that these improvements remained positive when stratified by acute vs. chronic conditions, BMI, and sex. For patients with low back pain, the total change in the RMQ score indicated a significant reduction in disability, with a mean reduction of 4.3 points (95% CI = 3.6-5.1), which is both clinically important and statistically significant. Patients with neck pain experienced a significant and clinically important reduction in neck-related disability, with a change of 6.8 points (95% CI = 5.4-8.1). The results for the NDI remained consistent regardless of

										the method used to impute missing data. Post hoc analyses showed that when stratified by chronicity, sex, or BMI, positive changes persisted for both the RMQ and NDI. Clinically significant improvements were observed in general health outcomes, with a total change in the PCS of 4.9 (95% CI = 3.8-6.0) and a total change in the MCS of 3.2 (95% CI = 2.0-4.3). Post hoc analyses indicated minimal variation in outcomes when stratified by acute vs. chronic conditions, sex, or BMI. A large majority of patients expressed satisfaction with their chiropractic care, with 78.8% reporting being "very satisfied" and 18.9% "satisfied."
Clinic Workload, the Quality of Staff Relationships and Diabetes Management in Community Health Centers Catering to Latino and Chinese Patients <i>Vargas Bustamante A. (2017)</i> USA	Cross-sectional 75%	Examine the association between primary care practice climate—specifically MCW, QSR, and diabetic patients' experiences of care—and diabetes outcomes in fourteen CHCs that serve a high proportion of low-income adult Latino and Chinese diabetic patients.	CHCs in North California serving low-income adult Latino and Chinese diabetic patients.	Clinicians and staff and patients of the 14 recruited CHCs. Patients were included if they were 18 years or older, had two or more visits to one of the participating clinics in 2011, and had a type 2 diabetes diagnosis or prescription. Sample size: Clinician/staff=249 Patients=907	Data source: patient health records (baseline characteristics) Tools: Clinician and staff workplace climate survey Patient Assessment of Chronic Illness Care (PACIC-11), Problem Areas in Diabetes (PAID-5) scale.	Manageable clinical workload Quality of staff relationship <u>Team composition:</u> - o Physician - o PA - o NP - o RN - o Medical assistant - o Care coordinator - o Clinic manager - o Licensed vocation nurse - o Non-clinical staff - o Other clinical staff	None	Process measures for blood sugar (HbA1c), cholesterol (LDL-C), and blood pressure (BP) testing were created [9]. A value of one was assigned to process measures for tests conducted at least once, and zero for tests that were not performed. Binary clinical outcome measures were derived from	<u>Patient-level:</u> - o Self-reported number of comorbidities - o Self-reported health status - o Education level - o Health insurance status (none, private, public) <u>Patient and team-level:</u> - o Race/ethnicity - o Language (Chinese, Spanish, English speaking) - o Sex - o Age <u>Organizational-level:</u> none measured	Patients from clinics with higher QSR were statistically more likely to be tested for HbA1c (OR=1.508, p<0.1), LDL-C (OR=1.633, p<0.05), and BP (OR=1.897, p<0.05). These patients also had significantly lower odds of having their cholesterol under control (LDL-C <130) (OR=0.730, p<0.05). Patients from clinics with both high QSR and high MCW had greater odds of being tested for HbA1c (OR=1.514, p<0.05—MCW only), LDL-C (OR=1.602, p<0.05—QSR only), and BP (OR=1.867, p<0.05—QSR only).

								patients' clinic records to indicate whether their results met the HEDIS guidelines for diabetes control. As a result, dichotomous measures were scored as one for acceptable control (HbA1c <8.0%, LDL-C <130 mg/dL, and BP <140/90 mmHg) and zero for poor control.		Patients from clinics with high QSR (OR=0.754, $p<0.05$) and high MCW (OR=0.841, $p<0.05$) had lower odds of having LDL-C under control. Patients' experiences with diabetes care (measured by PACIC-11, were more positive in clinics with a less manageable workload and poorer staff relationships.
Community Health Center Efficiency. The Impact of Organization Design and Local Context: The Case of Indonesia <i>Miharti S. (2022) Indonesia</i>	Cohort study 75%	Examine how variations in efficiency between CHCs can be explained by the alignment between their organizational design characteristics and the specific aspects of the communities they serve.	CHCs retrieved from 37 districts' health profiles in Indonesia drawn from reports published by the Department of Health of each district in 2011.	CHCs in 2011 retrieved from the Department of Health. No information on the number of patients and clinicians/staff included. Sample size=598 CHCs	Data source: CHC input, output, and context data for 37 districts' health profiles were sourced from reports published by the Department of Health of each district in 2011 (e.g., Health Profile of Tangerang Regency, accessed on April 27, 2014). Some reports were downloaded from the official Ministry of Health (MoH) website, while others were obtained from district websites. Basic data on CHCs were collected from the MoH's official website. This data includes details about the number and types of CHC health staff.	Health staff mix: The variety of health staff types working in a CHC, referred to as the staff mix. The range of services offered: ranges from 0 to 3 and reflects the presence of access to care (24-hour care), beds or inpatient care, and ambulatory services Type of providers: The first group includes clinical staff who directly provide care to patients. This group consists of individuals with higher and medium educational backgrounds, including physicians, dentists, specialists, midwives, and nurses. The second group includes non-clinical staff who indirectly contribute to patient care. This group consists of nutritionists, public health and environmental health workers, and pharmacists. As laboratories are not mandatory units in a CHC, the presence of laboratory staff is optional and categorized as an additional profession that a CHC may employ, alongside other health staff. <u>Team composition:</u> Doctors (physician, dentist, specialist)	None	Delivering primary healthcare and providing mother and infant care: Mother and infant care was delivered through services such as vaccinations, contraception care, and deliveries attended by healthcare staff. Vaccination and contraception users: The number of vaccinated infants, the number of active contraception users (females of childbearing age), and the number of deliveries attended by healthcare staff. Number of outpatient visits and health promotion	<u>Patient-level:</u> - Poverty rate (% of poor people in a CHC coverage area) - Level of remoteness (remote and none-remote) <u>Team-level:</u> none measured <u>Organizational-level:</u> - Population-size in a CHC service coverage area	The health staff mix ranges from 2 to 10, with an average of 7. Significant variations in CHC efficiency were observed, indicating that not all CHCs are equally effective at identifying the optimal design for efficient operation. In both remote and non-remote areas, CHCs with lower staff mix were associated with higher services delivered to mothers and infants such as higher vaccination rates and contraception users. In remote areas, the effect of staff mixes on the service delivery was higher versus non-remote areas ($P=0.05$).

					Timeline: data from 2011 collected retrospectively in August 2014	- Midwives - Nurses - Non-clinical staff (public health staff, environmental health staff) - Nutritionist - Pharmacist - Laboratory staff		activities.		
Community Health Center Residency Training: Improving Staffing, Service, and Quality <i>Chen C. (2021)</i> USA	Cross-sectional 87.5%	Examine effect of new Teaching Health Center (THC) residency programs on health center staffing, patient services, quality of care, and provider productivity.	Implementation of the Teaching Health Center (THC) residency programs into CHCs across 25 states in the US. THC are part of the Affordable Care Act, designed to specifically support the creation and expansion of community-based primary care residency programs. THCs are mainly family medicine programs situated in community health centers, rural health clinics, and tribal health centers.	In 2010, there were 22 health centers with new THC programs and 968 health centers without THC programs. CHCs participating in residency training were retrieved from the publicly available Health Resources and Services Administration (HRSA) THC-funded program list and through online searches and by reviewing residency program websites. Sample size=990 CHCs including 22 teaching centers and 968 non-teaching centers	Data source: HRSA Uniform Data System (US national database) Timeline: Data from 2010 were used for this study.	Addition of residents into the interprofessional team <u>Team composition:</u> - Physician - Advance practice clinician - Nurse - Medical staff - Mental health and substance use disorder specialist	New teaching health centers compared to non-teaching health centers	Staffing capacity: physician, advanced practice clinician, nurse, other medical, and mental health full-time equivalents (FTEs). Provider productivity: physician and total medical visits per FTE. Annual visit for specific services: numbers for mental health (covering alcohol-related disorders, other substance-related disorders, depression, and other mental health issues) and women's healthcare (including contraceptive management and deliveries). Quality outcomes: measures of controlled hypertension and controlled diabetes, low birth weight and early entry into prenatal care	<u>Patient and team-level:</u> none measured <u>Organizational-level:</u> Center and county-level characteristics: - Patient demographic - Rurality - County-level diabetes and HIV prevalence rate	THCs were associated with a statistically significant increase in the number of physician FTEs by 16.40 ($P<.01$) compared to non-THCs. The relative increases in advanced practice clinicians, nurses, mental health providers, and other medical providers' FTEs were also larger; however, the differences were not statistically significant. New THC programs were also associated with more total visits, with an increase of 33,415 visits per year relative to non-THCs at their 2010 baseline, though this was not statistically significant. Compared to non-THCs, there was a statistically significant decrease in the number of visits to new THCs. Health centers with new THCs were associated with having 425.3 fewer physician visits per FTE ($P<.01$) compared to non-THCs relative to the 2010 baseline. Centers with new THC programs were associated with a statistically significant higher volume of service visits for deliveries, with an increase of 231.0

								FQHC cost reports from the Healthcare Provider Cost Reporting Information System: identify resident FTEs in health centers with THC programs.		visits ($P<.05$) compared to non-THCs Visits for substance use disorder, alcohol treatment, depression, other mental health services, and contraception were all greater, but not statistically significant. Controlled hypertension, controlled diabetes, low birth weight, and early entry into prenatal care outcomes were all better in new THCs compared to non-THCs. Only early entry into prenatal care was statistically significant, with 4.90% ($P<.01$) more prenatal care patients entering early care in new THCs compared to non-THCs in 2018.
Community health centers employ diverse staffing patterns, which can provide productivity lessons for medical practice <i>Ku L. (2015)</i> USA	Cross-sectional 87.5%	Examine various staffing models in CHCs, the influence of staffing on productivity, the role of staff, and the factors that shape staffing patterns.	CHCs in the US in 2012 with complete data available and no missing data.	In 2012, there were 1,198 community health centers in the United States. Seven centers with missing data were excluded, leaving a final sample of 1,191 centers. Since community health centers may operate clinics at multiple locations, these 1,191 centers represented approximately 8,900 clinical sites. Sample size=1 191 CHCs Follow-up and attrition: NA	Data source: the 2012 annual reports submitted to the Uniform Data System (UDS) by all community health centers that received federal Section 330 grants. This included sites across all fifty states, the District of Columbia, and U.S. territories. Section 330 grants are the main federal funding provided to nonprofit community health centers by	Staffing pattern: Medical staff were categorized into four groups: physicians (including medical doctors and doctors of osteopathy, primarily in primary care specialties), advanced-practice staff (such as NP, PA, and certified nurse-midwives), nurses (registered, licensed vocational, and practical nurses), and other medical staff (e.g., medical assistants, nurse aides, and lab or radiology technicians). Staffing Clusters: The staffing patterns were grouped into four categories: 1) Typical, 2) High-advanced practice staff, 3) High nursing staff, and 4) High other medical staff. Concentration (%) of FTE Staff:	none	Productivity: number of weighted medical visits per center. The weight medical visits were calculated from an expenditure index for each site using the case-mix of diagnoses and the average expenditure per diagnosis.	<u>Patient-level:</u> - Medical patient status - Weighted medical visit - Mean weighted visits per FTE - Mean % of patients with limited English proficiency - Insurance status (insured/uninsured) - Medicaid status (yes/no) <u>Team-level:</u> - Mean % of staff (physician, APN, other medical) - Mean FTE medical staff - Mean physician per advance	Four primary staffing patterns were identified based on the distribution of medical staff across four categories. These patterns were: typical, high advanced-practice staff, high nursing staff, and high other medical staff. The typical group had a staff distribution closely aligned with the national average. Physicians consistently made the highest marginal contribution to productivity, with one additional physician associated with 2,994 additional visits. Nurses had the lowest marginal contribution (292 additional visits), though

					the Health Resources and Services Administration. Timeline: retrospective data from 2010 to 2012	County-level data from the Area Health Resource File was used to examine the concentration of primary care physicians, advanced-practice staff, and nurses per 1,000 county residents, as well as the percentage of the population in each county living below the federal poverty level. Rural/Urban Location: A rural-urban continuum code was included, ranging from 1 (for counties in metropolitan areas with populations over one million) to 9 (representing the most rural areas). NP Scope of Practice: The UDS data was combined with state-level information on the scope of practice for nurse practitioners, which defines the range of services they can provide without physician oversight. Nurse practitioners' scopes of practice were categorized as full (able to treat patients and prescribe medications without a physician's authorization), partial (able to treat but not prescribe), or restricted (requiring most actions to be authorized by a physician). <u>Team composition:</u> ○ Physician ○ Advance practice provider ○ Nurse ○ Other medical staff (medical assistant, nurse aides, laboratory and radiology technicians)			practice staff <u>Organizational-level:</u> ○ Number of CHC ○ % urban center	this result was not statistically significant. Location played a significant role in staffing. Rural centers tended to employ more advanced-practice staff and nurses, while employing fewer physicians and other medical staff compared to urban centers. The local availability of healthcare professionals influenced staffing. Centers in counties with higher concentrations of primary care physicians per 1,000 residents employed more physicians and fewer advanced-practice staff. Similarly, centers in counties with more advanced-practice staff per 1,000 residents had a higher proportion of such staff but fewer physicians. Higher concentrations of advanced-practice staff in a county were associated with fewer other medical staff. Counties with more nurses per 1,000 residents had centers employing more physicians and fewer advanced-practice staff, though nurse concentrations did not significantly impact the share of nurses in staffing. State laws governing NP scope of practice also influenced staffing patterns. In states with
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										full NP scope of practice (allowing NPs to treat and prescribe without physician oversight), centers employed slightly fewer physicians and more advanced-practice staff. Full scope of practice was also linked to fewer non-advanced-practice nurses but a greater number of other medical staff. Staffing was also shaped by patient demographics and caseloads. Centers with larger patient volumes had more other medical staff but fewer advanced-practice staff. Those with higher proportions of uninsured patients employed fewer physicians and more advanced-practice staff. Medicaid patient percentages did not significantly influence staffing, but a greater share of patients with limited English proficiency was associated with increased use of other medical staff.
Cost-effectiveness of on-site versus off-site collaborative care for depression in rural FQHCs (economic evaluation) Pyne J. (2015) USA	RCT and economic evaluation Pyne: 90.9% Fortney: 61.53%	Compare the economic effectiveness of two alternative approaches, the telemedicine collaborative care approach (TBCC) and the practice-based collaborative care approach (PBCC), to delivering depression care	A total of 5 FQHCs located in Arkansas, US were enrolled in the study. These FQHCs employed between 1.3 and 9.7 full-time-equivalent primary care (PC) physicians, served between 5,362 and 13,050 unique PC patients, and	Between 2007 and 2009, a total of 19,285 patients were screened for depression. Of these, 2,863 (15%) screened positive with a Patient Health Questionnaire-9 (PHQ-9) score of ≥ 10, and 364 patients were enrolled in the study. Patients with schizophrenia,	Data source: Data were retrieved from the Arkansas FQHCs and data related to costs were retrieved from the national cost estimate. Tools: Symptom Checklist-20 (SCL-20).	Telemedicine based collaborative care (TBCC) includes an on-site and off-site part to the approach. On-site part: TBCC involved a team of five types of providers: on-site primary care (PC) providers and off-site specialists, including a depression care manager (DCM, a registered nurse), a clinical pharmacist (Pharm.D.), a psychologist (Ph.D.), and a	Practice-based collaborative care (PBCC) involved two types of providers: on-site primary care (PC) providers and on-site nurse depression care managers (DCMs). Each clinic employed a part-time DCM (half-time)	Depression symptoms Depression-free days (DFD) Quality-adjusted life years (QALYs) Fidelity measures: number of encounters with depression care	<u>Patient-level:</u> - Residential zip code - Urban/rural location <u>Team and organizational-level:</u> none measured	TBCC group experienced a significantly greater treatment response, higher odds of remission, and greater reductions in depression severity over time compared to the PBCC group, as measured by the SCL-20. While no statistically significant differences in healthcare costs were

Practice-Based Versus Telemedicine-Based Collaborative Care for Depression in Rural Federally Qualified Health Centers: A Pragmatic Randomized Comparative Effectiveness Trial (main findings) Fortney J. (2013)	management in Federally Qualified Health Centers (FQHCs). The practice-based collaborative care (PBCC), aimed to improve depression outcomes by leveraging local providers. The TBCC utilized remote specialists to support local primary care (PCP) providers. Compare the process and clinical outcomes of patients randomly assigned to either telemedicine-based collaborative care or practice-based collaborative care in small, remote primary care clinics without on-site mental health specialists.	operated between one and six clinics across multiple locations. None of the participating clinic sites had an on-site mental health specialist.	bipolar disorder, or acute suicidal ideation were excluded. Sample size=364	Medical Outcomes Study 12-Item Short-Form Survey (SF-12) results to derive QALYs. Quality of Well-Being Scale (QWB) to calculate QALYs. Demographic data: The following tools were used to collect patient characteristics: Depression Outcomes Module, the Mini International Neuropsychiatric Interview, the Duke Social Support and Stress Scale, the Quality Improvement for Depression Treatment Acceptability Scale, and the Depression Beliefs Inventory Tools: Depression Outcomes Modules Mini International Neuropsychiatric Interview Duke Social support and Stress Scale Quality Improvement for Depression Treatment Acceptability Scale	psychiatrist (M.D.). The off-site team, funded by the study, was based at the University of Arkansas for Medical Sciences (UAMS) Off-site part: All remote patient interactions were conducted via telephone, following the established protocol. The depression care manager met weekly with the psychiatrist to review new patients and those not responding to treatment, preparing progress notes with stepped-care treatment recommendations. These notes were faxed to the FQHC for the PC providers to implement. If a patient did not respond to the initial antidepressant, the off-site pharmacist conducted a medication history and provided medication management recommendations as needed. For patients who did not respond to two medication trials, a psychiatry consultation was arranged via interactive video. Additionally, patients could access cognitive-behavioral therapy through interactive video at any time. <u>Team composition:</u> On-site: - Primary care provider Off-site: - Depression care manager (registered nurse) - Clinical pharmacist - Psychologist - Psychiatrist	funded by the study. The DCMs underwent one day of training in depression care management, were provided with a care manager training manual, and had access to a web-based decision support system. DCM encounters were conducted either in person or by telephone, based on patient preference. The initial session included PHQ-9 symptom monitoring, patient education, self-management behavioral activation, barrier identification and resolution, and the establishment of self-management goals, such as planning physical, rewarding, and social activities. Follow-up sessions focused on monitoring symptoms using the PHQ-9, assessing medication adherence and side effects, and encouraging participation in planned self-management activities. PBCC DCMs did not receive supervision from a mental health	managers that included documented Patient Health Questionnaire scores; self-management activities; adherence to antidepressant treatment and side effect assessments for those prescribed antidepressants; and adherence to counseling for those referred to psychotherapy. Depression care manager fidelity was also assessed through patient self-reports during 6- and 12-month telephone interviews, which addressed education, self-management, symptom monitoring, adherence monitoring, and collaboration among providers. To assess fidelity to the stepped care protocol, we examined the proportion of patients in the telemedicine group who, after failing at least one medication trial, had a telephone consultation with a pharmacist, as well as the proportion who, after failing two medication trials,	found between the groups, the total cost per patient was significantly higher for TBCC compared to PBCC due to the higher fixed and variable costs associated with TBCC. The unadjusted average incremental intervention cost (difference between fixed and variable costs of TBCC and PBCC) was \$1,132 (\$376 + \$756). In the base case analysis, the adjusted total cost was significantly greater for TBCC than PBCC (b = 1,146, 95% CI = 396–1,897, P = .003). The adjusted incremental cost of TBCC in the base case was \$1,146, which aligns with the incremental costs reported for other collaborative care interventions for depression (\$389 to \$1,772 per capita, adjusted to 2009 dollars). Tele video equipment and T1-line charges accounted for 50% of the per-capita direct costs of TBCC. The results showed that TBCC was both more effective and more cost-effective compared to PBCC. The incremental cost-effectiveness of TBCC was \$10.78 per depression-free day, which is lower than the amount that depressed patients reported being willing to pay for an additional depression-free day (\$14.40, adjusted to 2009
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					Depression Health Beliefs Inventory 20-item Hopkins Symptoms Checklist 12-items Short-Forum Health Survey Quality of Well-Being Scale Consumer Assessment of Healthcare Providers and Systems Timeline: data collected at baseline, 6, 12 and 18 months after the beginning of the interventions.	specialist. However, patients could be referred to off-site specialists, such as community mental health centers, if necessary. Progress notes were recorded in the patients' paper medical records. The intervention was provided for up to 12 months. <u>Team composition:</u> On-site: ○ Depression care manager (registered nurse) ○ Primary care provider	had a telepsychiatry consultation. Additionally, we examined the proportion of patients in the telemedicine group who attended at least one tele psychotherapy session and completed at least eight sessions. Clinical outcomes: Depression severity, self-reported health status (physical and mental health), satisfaction with care	dollars). A significant main effect for group was observed for both response (odds ratio = 7.74) and remission (odds ratio = 12.69), with patients in the telemedicine-based group showing better outcomes. A significant group-by-time interaction effect was found for the HSCL score ($\chi^2 = 40.51$, $df = 3$, $p < 0.001$), indicating greater reductions in depression severity among patients in the telemedicine-based group. A significant interaction effect was observed for the mental health composite scale score ($\chi^2 = 11.46$, $df = 3$, $P = 0.01$) and Quality of Well-Being score ($\chi^2 = 6.55$, $df = 3$, $P = 0.09$), with more substantial improvements in these measures for the telemedicine-based group. No significant group-by-time interaction effect was found for the physical health composite scale score, though this result is likely due to an artifact from the orthogonal factor rotation and negative weights used in scoring the physical and mental health composite
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										scales.
										While there were no significant group differences in satisfaction at baseline, the omnibus test for group-by-time interactions was significant at the 0.10 level ($\chi^2 = 6.69$, $df = 3$, $P = 0.08$), indicating higher satisfaction in the telemedicine-based group.
Diabetes management at community health centers: examining association with patient and regional characteristics, efficiency, and staffing patterns <i>Thorson M. (2020)</i> USA	Economic evaluation 54.59%	Examine the association between a higher proportion of behavioral health and advanced-practice professionals on staff will be associated with improved diabetes outcomes.	The study focused on FQHCs funded under Section 330 of the Public Health Service Act, which mandates serving all residents in the center's service area regardless of their ability to pay.	This study excluded 44 FQHCs located outside the 50 states or serving fewer than 100 patients. Also, 79 additional centers funded exclusively under Sections 330(g), (h), or (i)—which serve specific subpopulations such as migrant workers, homeless individuals, or public housing residents—were excluded. Another, 18 CHCs with missing data on staffing FTEs were excluded, resulting in a final analytic sample of 1,234 CHCs for latent class analysis (LCA). For the Data Envelopment Analysis (DEA), five CHCs were further excluded due to missing data on the percentage of adult patients with uncontrolled diabetes ($HbA1c > 9\%$), leaving a final DEA sample of 1,229	Data source: Data were sourced from multiple levels, including CHC-specific and regional information at the zip-code tabulation area (ZCTA) level. Health center data were derived from the 2015 Uniform Data System (UDS), which includes details on patient demographics, health measures, and costs for all 1,375 federally qualified health centers (FQHCs; HRSA, 2015). Staffing data for each FQHC, categorized by major service area, were obtained from the UDS Mapper	Staffing pattern: % of FTE allocated across six staffing categories: 1) physician, 2) advanced practice professionals (NP, PA, certified nurse-midwives), 3) nursing and medical/technical staff, 4) enabling services (transportation, social services), 5) behavioral health staff (mental health and substance use), 6) vision and dental staff. <u>Team composition:</u> - Physician - Advanced practice providers - Nursing staff - Medical staff - Enabling services - Behavioral staff - Vision and dental staff	None	Number of patients at a CHC with uncontrolled diabetes: HbA1C level of 9% or higher or being a diabetes patient who had not received an HbA1C test in the past year	<u>Patient-level:</u> - Race (% black, Asian/pacific islander, Hispanic, American Indian) - Poverty (% of patients at or below 100% of the federal poverty level) - Insurance status (% uninsured, covered by Medicaid, enrolled in Medicare) - Average number of medical visits per patient <u>Team-level:</u> none measured <u>Organizational-level:</u> % population living in a urban areas % of people living in poverty (at or below 100% federal poverty line) % identified as non-white % classified obese ($BMI > 30$) % without access to a usual source	After adjusting for patient and organizational characteristics, FQHCs with high levels of staff in the category "behavioral health staff" had significantly lower odds of diabetes patients with uncontrolled diabetes compared to other staffing classes. FQHCs with high levels of staff in the category "nursing", "advanced practice professionals" and "vision and dental staff" were associated with 19% and 33% higher odds, respectively, of having diabetes patients with uncontrolled diabetes relative to the staff category of "behavioral health".

				CHCs. Sample size= Latent class analysis=1234; Development envelopment analysis=1229	website Regional-level information was gathered from the U.S. Census American Community Survey (ACS; 2010–2014) and the Behavioral Risk Factor Surveillance System (BRFSS; 2009–2012). To link regional data with each FQHCs service area, ZCTA-level data were averaged across all ZCTAs served by the FQHCs. Timeline: data retrieved from 2015				- of medical care % adult reporting a diabetes diagnosis - Centers for Medicare and Medicaid Services Regions	
Do years of experience with electronic health records matter for productivity in community health centers? <i>Frogner B. (2017)</i> USA	Cross-sectional 75%	Assess the relationship between productivity and years of experience with electronic health records (EHRs). Analyzed how the staffing mix changed with varying levels of EHR experience and investigated whether the impact of EHR experience on staffing productivity differed over time.	CHCs in the US includes all Section 330 grantees. The Health Resources and Services Administration (HRSA) of the U.S. Department of Health and Human Services defines health centers as entities that serve medically underserved areas or specific medically underserved populations.	CHCs in the US Unit of analysis: CHCs Sample size=1198 CHCs	Data source: The primary data source for our study was the 2012 Uniform Data System (UDS), an annual reporting system for all health centers receiving Health Center Grant funds under Section 330. This includes centers located in the 50 states, the District of Columbia, and U.S. territories Tool: The UDS data was supplemented	All CHCs were grouped into one of seven categories based on their years of EHR experience as of 2012: (1) no live EHR, (2) less than 1 year (adopted in 2012), (3) 1–2 years (adopted between 2010 and 2011), (4) 3–4 years (adopted between 2008 and 2009), (5) 5–6 years (adopted between 2006 and 2007), (6) 7 or more years (adopted in 2005 or earlier), and (7) unknown year of adoption. CHCs were grouped by the EHR vendor reported in 2012. Centers were assigned to one of the three most used vendors, which accounted for over half of the sample: NextGen, eClinicalWorks, and GE Centricity. All remaining CHCs, which used 15 different EHR vendors, were grouped into a category labeled "all other systems."	None	Productivity: total number of medical visits per center in 2012, adjusted for service intensity. To account for service intensity, the weight index was calculated from the average expenditures for an office-based visit across the 26 International Classification of Diseases, Ninth Revision (ICD-9) diagnostic categories reported in the UDS.	<u>Patient and team-level:</u> none measured <u>Organizational-level:</u> - Size of the CHC (whether the CHC served more than the median number of patients for medical visits) - Rural location - Capital input: % of expenses allocated to facility and nonclinical support services (cost for management and support staff, fiscal and billing staff, facility staff, patient services support and staff)	CHCs with more years of EHR experience had a higher number of medical staff FTEs, although there was significant variation within each "years of experience" category. The proportion of advanced practice providers (e.g., NPs and PAs) was consistent across all levels of EHR experience. CHCs with EHRs had a higher share of other medical personnel FTEs (6–10 percentage points higher) compared to CHCs without EHRs. CHCs with EHRs had lower shares of physician FTEs (1–4 percentage points) and

					with the "Readiness for Meaningful Use and HIT and Patient-Centered Medical Home (PCMH) Recognition Survey,". The survey was conducted between December 2010 and February 2011. Timeline: Data were retrieved in 2012	<u>Team composition</u> (number of FTE in the following categories): ○ Physician ○ Advanced practice provider (NP, PA, certified nurse, midwife) ○ Nurse (RN, licenses practice nurses, vocational nurse) ○ Other medical staff (medical assistant, nurse aides, staff involved in quality assurance and/or EHR programs)				nurse FTEs (4–7 percentage points) compared to those without EHRs. CHCs without EHRs had the highest mean and median medical visits per combined physician, NP, and PA FTE despite having smaller staff and fewer total visits. All staff types significantly contributed to CHC productivity across all "years of experience" categories and in CHCs without EHRs. The only exception was that nurses (excluding NPs) did not contribute significantly to productivity in CHCs with 1–2 years of EHR experience. In CHCs with EHRs, physicians contributed significantly more to productivity than those in CHCs without EHRs. A 10% increase in physician FTEs resulted in nearly double the increase in visits in CHCs with EHRs (4% increase) compared to those without (2% increase). Physicians had the highest productivity contribution in CHCs with 3–4 years of EHR experience, where a 10% increase in physician FTEs was associated with a 5% increase in visits. Productivity
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										contributions were lowest in CHCs with 5+ years of EHR experience, though still higher than in CHCs without EHRs. Across all "years of experience" categories and CHCs without EHRs, no statistically significant differences were found in productivity contributions by advanced practice providers, nurses (excluding NPs), and other medical personnel. Nurses contributed the least to productivity in CHCs with 4 or fewer years of EHR experience, while their contributions in CHCs with 5+ years of EHR experience were comparable to those without EHRs. Other medical personnel consistently had low productivity contributions, while advanced practice providers maintained higher contributions. The productivity contributions of physicians, advanced practice providers, and other medical personnel did not vary significantly across EHR vendors. Nurses (excluding NPs) had the lowest productivity contribution in CHCs using NextGen compared to other
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										vendors. Including laboratory, x-ray, and IT personnel in the analysis showed no significant contribution of these staff types to the production of medical visits.
Does expanded state scope of practice for nurse practitioners and physician assistants increase primary care utilization in community health centers? <i>Park J. (2019)</i> USA	Cross-sectional 87.5%	This study investigates the impact of expanded scope of practice (SOP) for nurse practitioners (NPs) and physician assistants (PAs) on the number of visits conducted by primary care clinicians—NPs, PAs, and primary care physicians (PCPs)—in community health centers (CHCs) from 2009 to 2015.	CHCs in the district of Columbia, US	A total of 1,007 CHCs reported consistently throughout the study period. The sample was restricted to those located in the 50 states and the District of Columbia, resulting in 980 centers. An additional 241 CHCs were excluded due to missing data on key outcomes and/or covariates. After these exclusions, 739 CHCs remained for analysis. The baseline characteristics of the 739 CHCs included in the study were similar to those of the 980 eligible CHCs Sample size=739	Data source: the primary data source was the longitudinal data from the 2009–2015 Uniform Data System (UDS), which is collected and maintained by the Bureau of Primary Health Care under the Health Resources and Services Administration. The secondary data source was the Area Health Resources File, which provides county-level data on health service resources, the market supply of healthcare providers, and other population characteristics. For information on NP scope of practice (SOP) laws, data was retrieved from state statutes and regulations from 2009 to 2015. For PA SOP laws, the AAPA's Six Key Elements of State PA Laws was used.	Scope of practice of NP and PA NP: Using state statutes and regulations from 2009 to 2015, data were gathered on NP scope of practice (SOP) laws for each state, focusing on two main areas: physician involvement in treatment and diagnosis, and prescriptive authority. For each year, each state was classified into one of the following categories based on NP SOP laws: (1) most authority (full practice and prescription), (2) moderate authority (full practice only), and (3) least authority (restricted practice and prescription). During the study period, all 12 states (AL, CO, CT, DE, HI, MD, MA, MN, NV, ND, RI, and VT) expanded their NP SOP regulations. PA: The degree of state restrictions was assessed between 2009 and 2015 based on six key elements identified by the AAPA (2017), including licensure as a regulatory term, full prescriptive authority, whether a written agreement between the supervising physician and PA establishes SOP at the practice level, adaptable supervision requirements, whether cosignature requirements are determined at the practice level by the supervising physician, and the number of PAs a physician can supervise at one time. The more of these elements that were present, the more liberal the state law	None	Primary care utilization by each of the clinician type: number of severity-adjusted visits per full-time equivalent (FTE) primary care clinician annually	<u>Patient-level</u> (aggregated characteristics): - Age - Gender - Race/ethnicity - Insurance status - Limited English proficiency - Poverty - Median household income - Unemployment rate <u>Team-level:</u> None measured <u>Organizational-level:</u> - Support staff per clinician - EHR adoption - Patient-centered medical home status % of grant on total revenue - Size - Rural/urban location - Number of NP/PA/physicians per 1000 population	Across all CHCs, regardless of clinician type, there was no statistically significant change in the number of visits per full-time equivalent (FTE) clinician resulting from liberalizing either NP or PA scope of practice (SOP) laws. Statistically significant interactions between SOP laws and levels of support staff indicated that the number of visits per clinician type was more responsive to the level of support staff when the state regulatory environment became more favorable for NPs and PAs. Among CHCs with a high proportion of advanced practice staff, granting independent practice and prescriptive authority to NPs led to an increase in the number of visits attributed to each FTE NP (a marginal increase of 396 visits per FTE NP per year at $p < .1$), but only when the number of support staff per clinician increased. The number of visits per FTE PA per year

					Timeline: 2009 to 2015	for PA SOP. States were categorized into PA SOP laws from three levels of authority based on the number of elements present: (1) most authority (5–6 key elements), (2) moderate authority (3–4 key elements), and (3) least authority (0–2 key elements). <u>Team composition:</u> - Physician - NP - PA				decreased by 1,304 when NPs were granted practice authority ($p < .05$). The number of visits per FTE PCP per year also decreased by 705 when NPs were granted both practice and prescriptive authority ($p < .05$), and by 952 when NPs were granted practice authority ($p < .05$). States with greater PA SOP authority saw a significant decrease in NP visits (ranging from 755 to 959 visits per FTE NP per year) among CHCs with a high proportion of advanced practice staff. NP visits significantly increased (ranging from 295 to 415 visits per FTE NP per year) in CHCs with a high proportion of other medical staff.
Does the Patient-centered Medical Home Model Change Staffing and Utilization in the Community Health Centers? <i>Park J. (2018)</i> USA	Cross-sectional 62.5%	To examine the workforce changes occurring in community health centers that have attained patient-centered medical home model (PCMH) status and to evaluate how these changes are related to utilization, as indicated by the number of visits.	CHCs in the district of Columbia, US receiving the Federal Section 330 grants.	CHC in the district of Columbia, US. A total of 1,021 CHCs reported consistently during our study period. After further restricting the sample to those located in the 50 states and the District of Columbia, 994 CHCs were eligible and 750 of those were included in the final sample size. Sample size=750	Data source: Data were retrieved from five sources: (1) the Uniform Data System (UDS); (2) a roster of CHCs recognized as PCMHs through the HRSA PCMH; (3) the Readiness for Meaningful Use and Health Information Technology and PCMH Recognition Survey (referred to here as the	Adoption of the PCMH model. Practices seeking PCMH recognition must meet standards for organizing primary care, which emphasize enhanced access, patient-centered, comprehensive and coordinated care, and ongoing quality improvement. The primary explanatory variable of interest was whether the CHC adopted a PCMH each year, referred to as the “treatment.” This treatment occurred at various points between 2007 and 2013, with the timing differing across CHCs. In total, 450 CHCs became PCMHs, while 300 did not during the study period.	300 CHCs none-adopter of the PCMH model	The number of full-time equivalents (FTEs) in the following categories: A) PCPs; B) advanced practice staff; C) nurses; D) other medical staff; E) mental health/substance abuse; and F) enabling service staff (including case managers and health educators).	<u>Patient-level</u> (aggregated data): - Age - Sex - Race/ethnicity - Insurance status - English proficiency - Poverty <u>Team-level:</u> none measured <u>Organizational-level:</u> - Size % of grant over total revenue - EHR adoption ($<$ or $>$ 6 years) - Number of physician/NP/PA	After the adoption of the PCMH model, there were significant increases in the number of FTEs across all staffing categories, except for PCPs and nurses. CHCs that adopted the PCMH model saw an average increase of 0.53 FTE in advanced practice staff, which represented an 8.77% increase compared to an overall mean of 6.04 FTE ($P < 0.001$).

					"Readiness Survey"); (4) the Area Health Resources Files (AHRFs); and (5) state NP scope of practice laws. Data from the Readiness Survey, conducted between December 2010 and February 2011 by the Geiger Gibson Program in Community Health Policy at George Washington University, was used to determine the status of PCMH adoption prior to 2011. Timeline: 6-year period, between 2007 to 2013	<u>Team composition:</u> ○ PCP (family practitioner, general practitioner, internist, obstetrician/gynecologist, pediatrician) ○ Advanced practice staff (NP, PA, midwives) ○ Nurse (RN, CNS, licensed practice, vocational nurse) ○ Other medical staff (medical assistant, nurse aides, quality assurance, EHR staff, unlicensed internist and residents) ○ Mental health and substance abuse staff (psychiatrist, psychologist, social worker) ○ Enabling service staff (case manager, health educator)		Utilization was defined as the number of visits per center per year within each of the five staffing categories, as well as the total number of visits across all categories.	○ per 1000 populations ○ State laws on NP SOP	CHCs with the PCMH model also experienced significant increases in the number of other medical staff by 1.23 FTE (a 7.46% increase, $P = 0.001$), MH/SA staff by 0.73 FTE (a 17.63% increase, $P = 0.005$), and enabling service staff by 0.36 FTE (a 6.14% increase, $P = 0.079$). Overall, there was no significant change in the total number of visits per year with PCMH adoption. PCMH adoption was also significantly associated with an increase in visits by MH/SA staff (709 visits, a 16.08% increase, $P = 0.026$) and enabling service staff (537 visits, an 11.61% increase, $P = 0.009$). These changes, however, were not statistically significant over time. PCMH adoption had a significantly positive effect on marginal utilization only for advanced practice staff at the 0.05 level. Among PCMHs, an increase of 1 FTE in advanced practice staff was associated with an estimated increase of 539 visits per year (a 0.89% increase relative to an overall mean of 60,333 visits, $P = 0.039$).
Enhancing	Cross-	Compare the	Rural health	CHCs in the US	The primary data	Staffing was measured as full-	None	Number of	Patient-level:	In rural centers, each

staffing in rural community health centers can help improve behavioral health care <i>Han X. (2019)</i> USA	sectional 75%	staffing and services related to behavioral health in rural and urban community health centers over a more recent period (2013–17).	centers in the US that receive the 330-grant benefit from federal funding in multiple ways.	Sample size= 1384	source was the Uniform Data System (UDS) for the years 2013–17, provided by the Bureau of Primary Health Care, HRSA. The UDS data was combined with administrative data for 2013–17 from the HRSA Bureau of Health Workforce, which includes information on current NHSC participants and their practice sites, using the health center's grant number. Integrated county-level poverty and insurance data were collected from the Area Health Resources Files, another HRSA resource, by ZIP codes of the health center's headquarters. Timeline: 2013 to 2017	time equivalent (FTE) staff members who were regular employees of a health center. For simplicity, staffing analyses are based on FTEs. Behavioral health staff were categorized into two groups: mental health staff (including psychiatrists, licensed clinical psychologists, licensed clinical social workers, and other licensed mental health providers) and substance use staff. Other personnel, such as physicians, nurses, medical aides, and educators, may have assisted in delivering mental health or substance use services but were not counted as behavioral health staff. <u>Team composition:</u> - Behavioral staff - Physician - Nurse - Medical aides - educators		annual visits for behavioral health services at each center, including visits for substance use disorders (drug, alcohol, or tobacco), depression and other mood disorders, anxiety disorders, attention deficit and disruptive behavior disorders, and other mental health conditions. These visits were reported by each center to the UDS, using diagnosis codes from the ICD-9.	- Age - Race/ethnicity - Insurance type - Chronic condition - Managed care enrollment status <u>Organizational-level:</u> 330-grant type (community, migrant, homeless agent) - County poverty - County uninsurance rate - Total number of patients enrolled per center	additional staff FTE was significantly associated with an increase of 411 visits for substance use disorders, 539 visits for depression, 466 visits for anxiety, 151 visits for attention deficit and disruptive behavior disorders, and 300 visits for other mental health disorders annually. Rural centers had 54 more annual visits for substance use disorders per additional behavioral health staff member compared to urban centers. However, no significant difference in productivity levels was observed for other types of mental health visits between rural and urban centers. Each additional NHSC behavioral health staff FTE in rural centers was linked to 1,037 more visits for depression and 1,005 more visits for anxiety annually, which is roughly twice the contribution made by an additional NHSC staff FTE in urban centers (with a difference of 617 more visits for depression and 484 more visits for anxiety).
Improving access to hepatitis C treatment using clinical pharmacist services in an uninsured, at-risk population at a community health center <i>Fitzgerald L. (2021)</i>	Cohort study 45.45%	Evaluate a team-based approach implemented in a primary care medical home, incorporating clinical pharmacist services, to effectively treat hepatitis C virus (HCV) infection in an uninsured,	Lincoln Community health center in Durham NC. In 2019, LCHC served 36,361 patients, with over 67% living at or below 200% of the federal poverty level, 57% uninsured, and 1.1% experiencing	Individuals with HCV treated at the Lincoln CHC. During the first two years of the initiative (July 1, 2017, to June 30, 2019), 84 uninsured patients began treatment at LCHC using Patient Assistance Programs (PAP). Of the 84 patients, four (5%) did not	Data source: Laboratory blood sample: Prescribing providers typically ordered a repeat hepatitis C virus ribonucleic acid (HCV RNA) test approximately 4–6 weeks after treatment initiation. While	Addition of a clinical pharmacy service for HCV patients. Patients receiving treatment for HCV at the LCHC Health Care for the Homeless Clinic were seen by clinic staff as frequently as every two weeks, as telephone monitoring for this population often proved challenging. Additionally, clinic staff assisted in relaying medication refill information to patients upon request from the HCV pharmacist.	None	% of patient who achieved sustained virologic responses (SRV) among those who completed the 12-weeks post-treatment testing.	<u>Patient, team and organizational-level:</u> none measured	Of those who underwent 12-week lab monitoring, nearly all (65 out of 67, or 97%) achieved sustained virologic response (SVR-12) Two patients (3%) did not achieve a cure, as the virus was detected during the SVR-12 test. One treatment failure occurred because the patient did not complete

USA		at-risk population.	homelessness. The patient population was composed of 50% African American, 10% White, 8% identifying as other races, and 32% reporting more than one race, with 52% identifying as Hispanic.	complete treatment at LCHC. Two transferred their care to a correctional facility, and two discontinued treatments early (one before four weeks and one before eight weeks), with no follow-up testing after starting treatment. The final sample size consisted of 80 patients. Sample size=80 patients	not mandatory, this test helped verify treatment adherence and effectiveness while motivating patients to continue their therapy. A final HCV RNA test was conducted at least 12 weeks after the treatment end date (SVR-12) to confirm HCV cure. Patient health record Timeline: July 2017 to July 2019 (first two years of the initiative at the CHC)	<u>Team composition:</u> - Primary care provider (physician, NP, PA) - Patient assistance coordinator - Pharmacist - Care coordinator				the full course of treatment due to medication theft. The other case involved a patient who initially had no detectable virus four to six weeks after starting treatment but later tested positive at the SVR-12 test, possibly due to a resistant virus variant.
Influence of an interdisciplinary diabetes specialist team on short-term outcomes of diabetes at a community health center Katz G. (1998) USA	Cohort study 18.18%	Evaluate the impact of a one-year diabetes team intervention on short-term diabetes outcomes.	Government-funded neighborhood community health center in East Harlem, New York City, staffed by primary care physicians and allied healthcare workers. The center provides primary medical, gynecological, and podiatric care to community residents.	Patients served at the East Harlem CHC and referred by their PCP to the diabetes interprofessional team. Sample size=50 participants	Data source: After one-year, retrospective reviews of medical records were conducted for patients referred to the diabetes team. The data collected included demographic information, blood pressure, body mass index, the presence of diabetes-related complications, glycated hemoglobin levels, lipid profiles, changes in diabetes regimens, documentation of home glucose monitoring (HGM) practices,	The diabetes team, composed of a diabetologist, a bicultural certified diabetes nurse educator, and a nutritionist, conducted weekly clinics at the health center. They provided nutritional counseling, diabetes education, and management services in each patient's primary language. Patients met individually with all three team members. The diabetes nurse educator conducted a thorough assessment of each patient's diabetes knowledge and self-care skills, followed by education aligned with the American Diabetes Association (ADA) guidelines. The nutritionist offered dietary counseling, emphasizing ethnic foods, food models, and personalized meal plans. Each patient also consulted with the diabetologist, who gathered a comprehensive medical history and performed a physical	Baseline measure pre-intervention compared to one-year post-intervention	% of patient performing home-glucose monitoring (HGM) % of patient experiencing hypoglycemia % of patient management regimen including diet modification and usage of administered agent and insulin Glycosylated hemoglobin levels (mean +/- SD)	<u>Patient, team and organizational-level:</u> none measured	Following intervention by the diabetes specialist team, 33 patients (66%) were performing HGM, as documented through review of their records during clinic visits. This marked a significant increase in HGM practices ($P < 0.001$). Based on HGM records and reported symptoms, 14 patients (28%) were identified as having previously unrecognized hypoglycemic episodes. After adjustments to their diabetes treatment regimens by the diabetes team, hypoglycemia resolved in all but two patients (4%) with type 1 diabetes. Post-intervention, 5 patients (10%) managed their diabetes through

					and evidence of hypoglycemia.	examination. <u>Team composition:</u> - Diabetologist - Bicultural certified nurse educator - Nutritionist				dietary measures, 14 (28%) were on sulfonylurea therapy, and 31 (62%) were treated with insulin ($P < 0.05$). Mean glycosylated hemoglobin levels improved significantly in the 36 patients without previously unrecognized hypoglycemia. Patients with hypoglycemia initially had lower glycosylated hemoglobin levels, which tended to rise after their diabetes regimens were adjusted to prevent further hypoglycemic episodes. Among patients without hypoglycemia, mean glycosylated hemoglobin decreased from 12.1% at baseline to 10.8% after intervention by the diabetes team ($P = 0.025$)
Integrating a multidisciplinary pain team and chiropractic care in a community health center: an observational study of managing chronic spinal pain <i>Prater C. (2020)</i> USA	Pilot observational study 81.81%	Examine the impact of a multidisciplinary chronic pain team and chiropractic care to enhance pain management and provide patients with non-opioid treatment options.	Affinia Healthcare (AHC) is a Federally Qualified Health Center (FQHC) serving an underserved population in St. Louis, MO. With five clinical sites, AHC provides care to over 40,000 individuals each year.	Patient with chronic spinal pain (back and neck) diagnosed by their PCP and referred to one of the two interventions (multidisciplinary chronic pain team or chiropractic care). The sole exclusion criteria was an active cancer diagnosis. Sample size=42 Final sample size=35 Chiropractic team=20 Pain team=12 Both team=3	Tool: The Pain Disability Questionnaire (PDQ). Timeline: baseline measure Follow-up measure (6 to 21 months depending on the participants length of treatment, average of 11	Pain Team: The pain team was made up of a primary care physician (PCP), a behavioral health consultant, a clinical nurse, and a clinical pharmacist. The goal of the pain team was to reduce chronic pain, enhance psychosocial support, and decrease or gradually reduce opioid use for pain management. <u>Team composition:</u> - Primary care physician - Behavioral health consultant - Clinical nurse	Baseline compared to follow-up measured (mean follow-up occurred at 11 months after the baseline assessment (range from 6 to 21 months)	Function and disability affected by pain	<u>Patient-level:</u> - Age - Sex - Race - BMI - Insurance type - Preferred language - Presence of another pain diagnosis - Active mental health diagnosis (including substance abuse) - Diagnosis of hypertension and/or diabetes - Medication usage <u>Team-level:</u> none	The mean baseline PDQ score was 92.4 ± 6.1, and the mean follow-up PDQ score was 81.9 ± 7.7, resulting in a mean change of -10.6 (95% CI: 1.2 to -22.3, $P = 0.08$), where a negative value indicates improvement in pain scores. Univariate analysis showed that participation in the chiropractic team ($P < 0.01$), absence of another pain diagnosis ($P = 0.02$), and completing the follow-up

					months)	○ Clinical pharmacist Chiropractic Care: The chiropractic team consisted of faculty professors and students from Logan University in Chesterfield, MO, and on-site chiropractic care was made possible through an academic affiliation agreement. Chiropractic care included patient education, rehabilitation exercises, home care advice, and manual therapy techniques such as low or high-velocity, low-amplitude spinal manipulation, and myofascial treatment. The specific treatment plan for each patient was determined by the attending chiropractor in collaboration with the patient, based on their preferences. <u>Team composition:</u> ○ Chiropractor ○ Faculty professor ○ Students ○ Medical officer			measured <u>Organizational-level:</u> ○ Timing of the follow-up survey in relation to the 2019 covid-19 virus (before/after March 20 2020)	survey before COVID-19 ($P < 0.01$) were statistically significant predictors of better improvement in PDQ scores. However, after adjusting for all variables using multivariate linear regression, only COVID-19 and the type of intervention remained significant predictors of the outcome. The completion of the follow-up survey before COVID-19 ($P < 0.01$) and enrollment in the chiropractic team ($P = 0.01$) were associated with a greater improvement in PDQ scores following intervention. Both participants had successfully been weaned off opioids at follow-up. One participant experienced a slight increase in PDQ score (2 points), while the other had a more significant increase (31 points), with a mean increase of 16.5 points.
Optimal staffing in community health centers to improve quality of care Luo Q. (2021) USA	Cohort study 45.45%	Examine how the structure and deployment of primary care health professionals influence the delivery of quality care in Community Health Centers (CHCs).	CHCs in the US receiving the section 330 grant under the Public Health Service Act.	CHCs in the US Sample size=3 139 CHC year observation from 1178 CHCs	Data source: The primary data source for this study was the Uniform Data System (UDS) from 2014 to 2016. An additional key data source was the Internal Revenue Service (IRS) Form 990	Health workforce factors and capital expenditure. Labor inputs were represented by the number of Full-Time Equivalents (FTEs) for primary care providers (PCPs), physician assistants (PACs), nurses, other medical support staff, and administrative and enabling staff. Capital expenditure was measured in units of \$100,000, which included the rental and	None	Average % of patients in a CHC with well-controlled diabetes or hypertension. For diabetes patients, a well-controlled condition is defined as an A1C (blood	<u>Patient-level</u> (aggregated data): ○ Poverty rate ○ Uninsured rate ○ Area means income ○ Unemployment rate ○ % of female population ○ % black population ○ % Hispanic population	PCPs (0.260, $P = 0.055$) and APCs (0.252, $P = 0.062$) had similar direct marginal effects on quality, both of which were statistically significant. Nurses had a positive, but not statistically significant, direct effect on quality (0.076, $P = 0.277$), while other medical support staff

					nonprofit tax return database, which provided information on the capital of CHCs. This database offered detailed data on capital costs for nonprofit CHCs. Timeline: 2014 to 2016	depreciation costs of office buildings and equipment. <u>Team composition:</u> ○ Primary care physician ○ Advanced practice clinician (NP, PA) ○ Nurse ○ Other medical staff (e.g., medical assistant) ○ Administrative and enabling staff		sugar) level lower than 9 percent for a given year, as per UDS data. For hypertension patients, a well-controlled condition is defined as a blood pressure reading below 140/90 mm Hg. These outcome measures were calculated in UDS as the average proportion of diabetes patients with controlled A1C levels and the proportion of hypertension patients with controlled blood pressure. The final outcome measure used—the percentage of diabetic or hypertensive patients with their condition under control—had a mean of 63.91 percent during the study period. The second dependent variable was the total revenue of the CHC, defined as the sum of patient-	<u>Team-level:</u> none measured <u>Organizational-level:</u> ○ CHC wage index: overall cost of labor used by CHCs in the labor market ○ Prevalence of hypertension and diabetes ○ Number of behavioral health, dental health and laboratory FTE ○ Case-mix index ○ Number of delivered site	(−0.022, $P = 0.107$) and administrative and enabling staff (−0.64, $P = 0.107$) had negative, though not statistically significant, direct effects on quality. Capital investment showed a positive, but not statistically significant, effect on quality (0.033, $P = 0.663$). Both PCPs (0.319, $P = 0.015$) and APCs (0.299, $P = 0.022$) had similar and statistically significant total quality effects. However, the total quality effects of other factors were not statistically significant. For service volume, each additional PCP could perform 1,736 visits ($P < 0.001$), while an additional APC could perform 1,382 visits ($P < 0.001$). An additional nurse could increase visits by 268 ($P = 0.004$), and an additional administrative or enabling staff member could increase visits by 116 ($P = 0.048$). Each additional PCP could generate \$326,875 ($P < 0.001$), while an additional APC could generate \$183,914 ($P < 0.001$). An additional nurse could generate \$60,694 ($P < 0.001$), an additional other medical support staff member could generate \$38,580 ($P < 0.001$), and an additional administrative or enabling staff member
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								related revenue, grants, and contracts.		could generate \$53,118 (P < 0.001). One PCP is equivalent to 1.78 APCs in terms of overall productivity, as measured by revenue. However, due to the lower wage of APCs, hiring 1.78 APCs instead of 1 PCP would result in a 13.96% savings in salaries (\$100,538 * 1.78 - \$208,000) / \$208,000. At current wage levels and considering substitution effects, hiring one additional PCP is equivalent to hiring 1.77 APCs in terms of overall revenue productivity, generating a 14.5% salary savings.
Organizational Factors Associated with Disparities in Cervical and Colorectal Cancer Screening Rates in Community Health Centers <i>Chuang E. (2019)</i> USA	Cross-sectional 87.5%	Identify the organizational factors related to cervical and colorectal cancer screening rates among health centers funded by the Health Resources and Services Administration (HRSA).	Fully operational US CHCs who received Section 330 grant from the HRSA in 2015. CHCs were considered fully operational if they employed at least one FTE medical provider, one FTE administrative staff member and had a minimum of 5000 annual patient encounters. CHCs located in US territories were excluded.	CHCs in the US Sample size=984	Data source: Data for this study were obtained from the 2015 Uniform Data System (UDS) and the Area Health Resource File (AHRF). Timeline: 2015	% of health center revenue from various sources Net revenue per patient was included as an overall measure of health center financial health, calculated by subtracting total accrued costs from total revenue and dividing by the total number of patients seen. Organization size was defined as the total number of patients seen during the reporting year (in thousands) and serves as an indicator of health center capacity. Staffing ratios were used as measures of provider productivity and the capacity to offer preventive care during visits. The provider-to-patient ratio was calculated as the number of primary care provider full-time equivalents (FTEs) per 10,000 patients,	None	Cervical and cancer screening rate: appropriate colorectal cancer screening included having received a documented colonoscopy within the last 10 years, a flexible sigmoidoscopy within the last five years, or a fecal occult blood test (FOBT) or fecal immunochemical test (FIT) within the last year. For FOBT/FIT to qualify as appropriate screening, stool specimens had to be collected by the patient at home, and the test samples had to be	<u>Patient-level:</u> % of uninsured patient % homeless patient % of patient who were African American, Hispanic/Latino, Asian American, native Hawaiian or other pacific islander, American Indian or Alaska native <u>Team-level:</u> - Availability of the PCP: total number of PCP per 10 000 people within the county - Number of obstetrician-gynecologist per 10 000 people in the county - Number of gastroenterologists per 10 000 people	The % of revenue from public capitated managed care was positively linked to colorectal cancer screening rates (IRR 1.28, p<.05). Other revenue sources and net revenue per patient were not significantly associated with either cervical or colorectal cancer screening rates. Provider-to-patient ratios were significantly associated with higher screening rates for both cervical and colorectal cancer (IRR 1.01, p<.05 for cervical cancer screening and IRR 1.01, p<.05 for colorectal cancer screening). However, when controlling for other factors, support staff-to-provider ratios were not significantly associated with either screening

						while the support staff-to-provider ratio was the number of administrative and clinical support staff FTEs divided by the number of primary care provider FTEs. EHR status was included as an indicator of the technical infrastructure available to support guideline-concordant care, operationalized as a categorical variable reflecting the health center's participation in the Centers for Medicare & Medicaid Services (CMS) EHR Incentive Program: set to 0 if the center was not participating or was in the Adoption, Implementation, or Upgrade (AIU) stage; set to 1 if Stage 1 (i.e., basic EHR use for capturing and sharing clinical data); and set to 2 if in Stage 2 or 3 (i.e., using EHR to advance clinical processes or improve outcomes). <u>Team composition:</u> - Primary care provider - Clinical support staff - Administrative support staff		subsequently received and processed	in the county <u>Organizational-level:</u> Location in a non-metropolitan area	type. Electronic health record (EHR) status was significantly associated with higher cervical cancer screening rates (IRR 1.12, p<.01 for Stage 2 or 3) and colorectal cancer screening rates (IRR 1.09, p<.05 for Stage 1; IRR 1.10, p<.01 for Stage 2 or 3). Primary care provider availability was significantly linked to cancer screening rates (IRR 1.01, p<.05 for cervical cancer screening; IRR 1.02, p<.01 for colorectal cancer screening). When controlling for other factors, location in a non-metropolitan area and the availability of specialists were not significantly associated with colorectal cancer screening rates.
Provision of telemedicine services by community health centers <i>Shin P. (2014)</i> USA	Cross-sectional 25%	Describe the use of telemedicine services at federally qualified health centers and to assess whether health center characteristics differed based on the level of telemedicine services utilization.	All FQHCs in the US	Out of the 714 health centers that completed the readiness survey, 625 provided responses regarding the availability of telemedicine services (the responses of those who were "not sure" whether telemedicine was offered were excluded from the total of 625). Among these 625 health centers, 396 (63%) did not offer any	Data source: 2009 UDS database combined with the data from the survey completed in 2010-2011. Tool: The survey on Readiness for Meaningful Use (MU) of Health Information Technology and Patient-Centered Medical Home (PCMH)	The use of telemedicine by interprofessional team <u>Team composition:</u> - FTE per 10 000 patients: - Physician - Mid-level provider - Medical staff - Dental staff - Mental health and substance use staff - Enabling service staff	Telemedicine usage categories: A) Offered no telemedicine services, B) Offered one telemedicine service, C) Offered two or more telemedicine services.	Telemedicine usage (0, 1, 2 or more services provided) % of diabetic patient with HbA1c levels < 7%, % of diabetic patients HbA1C levels < 9% BP control among hypertensive	<u>Patient-level:</u> - Mean% of minority patient - Mean % of patient requiring translation service <u>Team-level:</u> - Meaningful use (MU) compliance status - EHR operation (full, partial, none) - Duration of EHR operation (less than a year ago, 1-2 years ago, 3+ years ago,	CHCs that did not offer telemedicine services were more likely to serve urban areas, while those providing two or more telemedicine services were significantly more likely to be in rural regions. Among CHCs offering two or more telemedicine services, 55% were in rural areas, 28% in urban areas, and 17% served both types of communities. In contrast, CHCs offering

				telemedicine services, while 229 (37%) provided some form of telemedicine services. Sample size=625 FQHCs	Recognition was conducted between December 2010 and February 2011. Timeline: December 2010 to February 2011.			patients Childhood immunization rate Birth weights Birth rate Pap test rate % of pregnant women with first prenatal visit in the first trimester	- o PCMH recognition - o Receiving technical assistance from a REC or sub-contractor - o <u>Organizational-level:</u> - o Location (rural, urban, both) - o Mean total patients, mean % Medicaid and Medicare - o Mean % uninsured - o Mean % elderly - o % of total revenues from Medicaid - o Mean Medicaid dollars per patient - o Received ARRA funding, - o Mean ARRA, NAP and IDS funds - o Mean ARRA CIP and FIP funds received - o % of total revenue from ARRA funds - o Mean ARRA funds per patient - o Mean ARRA funds per uninsured patient - o Mean state and local funds - o 10. % of total revenues from state and local fund - o Mean state and local funds per patient - o Mean states and local funds per uninsured patient - o % of total revenue from state and local funds	no telemedicine services were more likely to be in urban settings (47%), with 34.9% in rural areas and 18.2% in both urban and rural areas. CHCs providing two or more telemedicine services had higher staffing ratios for mid-level providers (5.2 FTEs per 10,000 patients) and medical personnel (25.9 FTEs per 10,000 patients). CHCs offering only one telemedicine service had the highest ratio of mental health providers (2.6 per 10,000 patients). An analysis of seven key quality of care measures from the UDS, including diabetes management, hypertension control, childhood immunizations, cervical cancer screening, birth weight, and prenatal care timing, revealed a significant difference only in the "percentage of diabetic patients with HbA1c levels below 7%" (indicating good diabetes control). CHCs with one telemedicine service reported the highest percentage at 42.2%. CHCs centers with two or more telemedicine services received significantly higher state and local funds per patient and per uninsured patient compared to centers that provided no
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										telemedicine services or only one telemedicine service.
Specialized care of overweight children in community health centers <i>Shin S. (2010)</i> USA	Cohort study 18.18%	Development of a patient-centered medical home intervention for children with obesity and its impact on chronic disease outcomes and management.	The Healthy Weight Clinic was first established at the Whittier Street Health Center in Boston, Massachusetts, in 2006, and has since been replicated at seven other community health centers across Massachusetts. These clinics are located in both urban and rural high-need areas, including Boston, Holyoke, Lawrence, Lowell, New Bedford, and Cape Cod.	From June 1, 2008, to August 31, 2009, a total of 402 children were served across eight sites. The first three clinics began the intervention in June 2008, while the next four started the program in July 2009. Whittier Street, the pilot site, launched its clinic in October 2006 but was included in the first clinical cohort for data analysis, which covered visits after June 2008. Out of 402 participants, only 174 had more than one visits at the clinic by August 2009. Sample size= 402 children	Data source: Data were collected via a web-based data collection and quality monitoring system to track key process and outcome measures, including BMI, diet, and physical activity. Tool: The Pediatric Obesity Assessment Tool Timeline: June 2008 to August 2009	At each site, multidisciplinary care is provided during designated clinic sessions by a team led by a primary care clinician with an interest in obesity, referred to as the "clinician champion." Patients participate in a series of one-hour individual visits, which include the clinician champion, a dietitian, and a case manager. The team members receive additional training in obesity management and health behavior change to become experts within the health center. Patients are referred to the Healthy Weight Clinic by their primary care providers. This model aligns with the core principles of the patient-centered medical home by including a clinician as the team leader and enhancing continuity and coordination of care through the provision of specialized services within the medical home. The clinician champion is responsible for the medical assessment of obesity, including reviewing laboratory results, family history, and other health conditions like sleep apnea. The dietitian evaluates sweetened beverage consumption, fruit and vegetable intake, and the patient or family's ability to recall what the patient ate in the past 24 hours. The case manager assesses sedentary behaviors, such as watching TV or playing video games, and physical activity levels. Children are typically seen every one to two months, for a total of six visits before graduating from the program. By having primary care providers refer patients to a specialized clinic within the same health system,	Patient with outcome compared with the total number of patients	Lifestyle modification: Reduced screen time and increase physical activity Diet modification: Decrease sweetened beverage and increase fruits and vegetable intake Decrease BMI	<u>Participant-level:</u> - Age - Sex - Ethnic group - Race - BMI <u>Team and organizational-level:</u> none measured	Out of all 402 children: 29.9% reduced screen time: and 45.5% increased physical activity: 32.2% decreased consumption of sweetened beverages and 33.3% increased fruit/vegetable intake. 79.8% mentioned having made any lifestyle and/or diet changes 50% decreased their BMI

						communication and care coordination are enhanced.				
						<u>Team composition:</u> ○ Clinician champion ○ Dietician ○ Case manager				
The impact of integrating medical assistants and community health workers on diabetes care management in community health center <i>Rodriguez H. (2018)</i> USA	RCT 38.4%	Assess the impact of integrating Community Health Workers (CHWs) and Medical Assistants (MAs) with expanded roles and responsibilities in managing diabetes care at Community Health Centers (CHCs) serving predominantly low-income Latino Spanish-speaking and Chinese patients.	16 Community Health Centers (CHC) were recruited across three counties in Northern California, all affiliated with the same regional community clinic association. These CHCs primarily served low-income Latino and Chinese patients and had bilingual, bicultural staff to provide care in the patients' preferred language. The CHC sites were randomized into three groups for a cluster-randomized trial. Using cluster analysis, practices were grouped into three sampling strata based on the most common combinations of three composite measures. Practices within each stratum were then randomly assigned to the Medical Assistant (MA) intervention group (n = 3), the Community Health Worker (CHW)	The patient analytic sample was identified as having diabetes based on ICD-9 codes 250, 357.2, 362.0, 366.41, and 648.0. A total of 167 randomly selected patients from each control practice and up to 400 patients from each intervention practice were included in the survey. Eligible patients had at least two visits with a diabetes diagnosis during the pre-intervention year (2011). Sample size=16 CHCs MA intervention=3 CHCs CHW intervention=3 CHCs Control group=10 CHCs 6,111 adult diabetes patients	Tools: Before site randomization (June–August 2011), practice surveys were conducted with CHC site directors (n = 15, response rate: 94%) and 281 adult primary care clinicians and staff from participating CHCs (response rate: 81%). Diabetes structural capabilities, 2) Primary care team functioning and 3) Practice size (mean number of clinicians/staffs. The patient experience survey targeting adult diabetic patients was conducted during two periods: the early intervention survey from July–August 2012 and the post-intervention survey with baseline respondents from July–August 2013. The surveys	Addition of MA and CHW in interprofessional teams. Both MAs and CHWs followed role-specific diabetes care management protocols developed by the clinics' quality improvement teams in collaboration with the regional community clinic association. MA intervention: The MA role focused on office-based panel management activities for diabetic patients, including maintaining the diabetes registry, regularly reviewing patient panels, following up on primary care clinician instructions, assisting with medication reconciliation, and identifying patients for appropriate interventions and referrals to community resources. MAs also received a brief module on health coaching strategies. CHW intervention: The CHW role emphasized broader patient health coaching skills, such as helping patients set agendas for clinician visits, ensuring patients understood clinician instructions, assessing agreement with care plans, supporting healthy behavior changes, and assisting with medication understanding and adherence. CHWs also received a less extensive module on panel management activities The intervention period lasted from January to December 2012, though all practices continued using the team-based models through 2013. Early intervention took place from January to February 2012	CHCs in the control group did not implement significant changes to their diabetes care processes during the study period, as confirmed through key informant interviews. However, most control CHC sites had some support in place for pre-visit ordering of routine services and follow-up to ensure these services were received, though this support was not specific to patients with diabetes.	HbA1C testing HbA1C controlled below 8% LDL-C tested LDL-C testing below 100mg/dl BP controlled below 140/90 mmHg Patients experience survey: A) Patient experience of care, B) Communication, C) Access to care, D) patient assessment of chronic illness, E) problems areas in diabetes, F) hypoglycemic events (% of reporting an event in the past 4 weeks), G) patient symptoms	<u>Patient-level:</u> ○ Age ○ Sex ○ Race/ethnicity ○ Language (Asian, Latino, other English speaker) ○ % of people uninsured ○ BMI (normal, overweight, obese 1 and 2, obese 3) ○ Comorbidities (cardiovascular, respiratory, mental health) ○ Number of CHCs visits during the study period <u>Team-level:</u> none measured <u>Organizational-level:</u> ○ CHC site	Patients in the CHW intervention group showed an 18.5%-point improvement in annual glycated hemoglobin (HbA1c) testing ($p < 0.001$). However, this improvement was not significantly different from the changes observed in the CHW control group. LDL-C testing, improved significantly more over time in the CHW control group compared to the CHW intervention group (9.8% points vs. 0%, $p < 0.01$). For the Medical Assistant (MA) intervention group, patients showed an 8.4%-point improvement in LDL-C control ($p < 0.05$) over time, but this improvement was not significantly different from the changes in the MA control group. HbA1c testing improved more for the MA control group compared to the MA intervention group (5.8% points vs. 1.0%-point, $p < 0.05$). No other differential changes were observed over time for other diabetes care processes or intermediate outcomes between the MA intervention and

			intervention group (n = 3), or the control group (n = 10).		covered care experiences over the prior six months. .A blood sample to measure hypoglycemia was included, including measures of the HbA1c and LDL-C.	<u>Team composition:</u> - Primary care physician - Nurse - MA or CHW				control groups. Patients in the MA intervention group reported improved chronic care experiences, as indicated by a 7.5-point improvement on the PACIC-11 ($p < 0.001$), with significantly greater improvement compared to control group patients ($P < 0.05$). Control group patients reported a reduction in symptoms of diabetes-related emotional distress (PAID-5) over time ($\beta = 3.4$, $P < 0.05$), but the improvement was no different from the changes observed in the MA or CHW intervention groups. Self-reported hypoglycemic events did not show any differential changes over time across the study arms.
The Role of the National Health Service Corps Clinicians in Enhancing Staffing and Patient Care Capacity in Community Health Centers <i>Han X. (2019)</i> USA	Cohort study 63.63%	Assess whether National Health Service Corps (NHSC) clinicians complement non-NHSC clinicians in Community Health Centers (CHCs) and compare their productivity, measured as visits per staff, to that of non-NHSC clinicians across primary	CHCs in the USA: 1,169 health centers that consistently reported to the Uniform Data System (UDS) over a four-year period were included. Health centers that began or ceased operations during the study period (232 centers) were excluded, as their staffing levels or visit	CHCs in the US reporting to the Uniform Data System Sample size=4096 center-year observation from 1024 CHCs	Data source: Data for this study were sourced from the 2013 to 2016 Bureau of Primary Health Care Uniform Data System (UDS) and Bureau of Health Workforce NHSC administrative records, both provided by the Health Resources and Services	The number of full-time equivalent (FTE) staff for both NHSC and non-NHSC clinicians across primary care, dental care, and mental health care at each health center. NHSC clinicians were defined as active participants in the NHSC federal loan repayment or scholarship program during the study period. <u>Team composition:</u> - Family physician - General practitioner	None	Number of non-NHSC clinician FTEs Number of annual patient visits in primary, dental and mental health care	<u>Patient-level:</u> - Race/ethnicity (Hispanic, non-Hispanic, black, etc.) - Income below 100% of the federal poverty level - Insurance status <u>Team-level:</u> - Number of administrative staff FTE - Local provider to population ratio	For every additional NHSC staff FTE, there was a significant increase of 0.72 (95% CI, 0.02–0.32) non-NHSC staff FTEs in mental health care. An increased NHSC staffing was linked to a significant growth of 0.04 (95% CI, 0.01–0.07) non-NHSC primary care staff FTEs in CHCs with more severe

		care, dental care, and mental health services.	volumes could only increase or decrease due to their entry or exit. The panel data was then merged with the NHSC staffing data based on grant numbers.		Administration (HRSA). State-level data on mental health provider employment data from the Bureau of Labor Statistics Occupational Employment Statistics were also used for this study. Timeline: 2013 to 2016	○ Internist ○ Obstetrician/gynecologist ○ NP ○ PA ○ Certified nurse midwives ○ Dental care (dentist, dental hygienist) ○ Mental health clinician (psychiatrist, substance abuse, etc.)			<u>Organizational-level:</u> ○ Rural or urban status ○ Type of federal grant received ○ % of health center patient who were aged 65 and above ○ County poverty rate ○ Year (2013 to 2016)	staffing shortages. No significant associations were found between NHSC and non-NHSC staffing in dental care. Each additional NHSC staff FTE was associated with an increase of 2,216 (95% CI, 934–3,497) primary care visits, 2,802 (95% CI, 2,019–3,585) dental care visits, and 1,296 (95% CI, 979–1,613) mental health visits in CHCs. Post-estimation tests revealed no significant difference in adjusted primary care visits between NHSC and non-NHSC clinicians, despite a slightly higher number for NHSC clinicians (difference = 194, 95% CI, –848 to 1,235). NHSC clinicians had significantly more adjusted visits in dental (difference = 992, 95% CI, 45–1,939) and mental health care (difference = 423, 95% CI, 59–788) compared to non-NHSC clinicians.
Association of chiropractic care with receiving an opioid prescription for noncancer spinal pain within a Canadian community	Sequential mixed-method (result paper #1 and #2) Pilot study and economic evaluation	Investigate the association between receiving chiropractic care and the initiation of opioid prescriptions among adult patients with noncancer	Lang CHC in Ontario Canada. The Langs CHC is situated in Cambridge, Ontario, Canada,	The study included adult patients who had two or more visits related to a diagnosis of back or neck pain not associated with cancer between January 1, 2014 (the launch date of the CHC's chiropractic	Data source: EHR data from the Lang CHC. Prescribed opioid dosage, were retrieved from the Narcotics Monitoring	The exposure of interest was the addition of chiropractic care to ongoing GP/NP care compared to ongoing GP/NP care alone. Recipients of chiropractic care were defined as patients who had at least one appointment for GP/NP care and at least one appointment for chiropractic care during the seven-year	Nonrecipient: received 2 or more appointments with their GP/NP alone For the pilot study, measures pre- and post-intervention were	Time to first opioid prescription: prescription between the time of the index (first visit with a chiropractor) and last visit dates for a noncancer spinal pain	<u>Patient-level:</u> ○ Age ○ Sex ○ Smoking status ○ Obesity ○ Depression ○ Anxiety ○ Fibromyalgia ○ Diabetes ○ Cardiovascular	Of the 945 patients included in the analysis, 19% (183 patients) received chiropractic services. Among these, 48% (87 patients) received chiropractic care within 30 days of their index visit.

health center: a mixed analysis (2022) <i>Result paper #1</i> The association between chiropractic integration in an Ontario community health centre and continued prescription opioid use for chronic non-cancer spinal pain: a sequential explanatory mixed method study (2022) <i>Result paper #2</i> Association of chiropractic integration in an Ontario community health centre with prescription opioid use for chronic non-cancer pain: a mixed methods study protocol (2021) <i>Study protocol</i> Chiropractic integration within a community health centre: a cost description and partial analysis of cost-utility from the perspective of the institution (2019) <i>Pilot study</i> Management of Back Pain-related Disorders in a	Service evaluation, cross-sectional pre and post study Emary 2022: 81.81% Emary 2022: 81.81% Emary 2019: 72.72% Emary 2017: 62.5%	spinal pain in a Canadian community health center. Examine if the addition of chiropractic care to standard medical care is cost-effective for adult patients with back pain in a CHC setting from the perspective of the healthcare institution. Describe a chiropractic service for back pain patients integrated within a multidisciplinary primary care CHC setting.	a mid-sized urban municipality with a population of approximately 130,000, located 82 km southwest of Toronto. The center serves local communities and vulnerable populations characterized by high unemployment rates, multiple comorbidities, and musculoskeletal disorders, which are often managed with prescription opioids. The pilot project was conducted in 2017.	program), and December 31, 2020. Patients with contraindications to chiropractic treatment, such as fractures, infections, inflammatory arthritis, or cauda equina syndrome, were excluded. In total, 1166 participants were identified of which 945 met the inclusion criteria. Sample size=945 participants For the pilot study, 95 participants were recruited of which 45 completed both the initial and follow-up assessment (47.4%). Sample size=45 participants Sample size=93 participants	System database of the ICES. Tools: EQ-5D (3L version questionnaire). Timeline: January 1st 2014 (launch of the program) to December 31st 2020 (6 years period). Timeline: The pilot study was held during the first two years of the program (January 2014 to January 2016). For each participant, questionnaires were administered at baseline and at 90-days follow-up. Complementary qualitative data collection Tools: A) The Bourne mouth Questionnaire (BQ), B) the Bothersomeness Questionnaire, C) the EuroQol 5-Domain (EQ-5D) scale to measure general health status. Timeline: January 2014 to January 2016. For each participant, the	study period. Eligibility for chiropractic services required a referral from a GP/NP, ensuring that all chiropractic care recipients were also receiving ongoing GP/NP care. The program is staffed on a rotating basis by a team of three to four chiropractors and is accessible only to CHC-registered patients through a referral from their GP. Treatment sessions are evidence-based and include spinal manipulation, soft-tissue therapy, patient education, reassurance, and home-care advice, such as icing, stretching, and strengthening exercises. Most patients are discharged after three months or a maximum of 12 visits, although some continue treatment to manage flare-ups or chronic pain. <u>Team composition:</u> - GP/NP - Nurse - Dietician - Social worker - Physical therapist - Massage therapist - Chiropractor	compared For the service evaluation study, measures pre- and post-intervention were compared.	diagnosis were included Opioid prescription including: opioid fills, refills (measured in 30-day equivalents), and dosage at baseline, 3, 6, 9, and 12 months. These outcomes were tracked for up to 12 months from the date of the first opioid prescription following the patient's index visit for chronic non-cancer spinal pain. For patients who received chiropractic care, their index visit was defined as their first chiropractic appointment to ensure temporality. Bournemouth questionnaire score Patient global impression of change (PGIC) Patient satisfaction EQ-5D score Cost-analysis Back pain General health status Improvement between baseline and discharge Patient satisfaction Medication usage	disease Frequency of healthcare visits Pilot study (self-reported) and service evaluation study - Age - Sex - Smoking status - In paid employment (yes/no) - Time-off work for current pain episode - Similar or same condition in the past - Pain free for a whole 6 months - Duration of the current pain episode - Constant pain - Limited movement - Limited usual activities - Medication intake - Other consultations with clinicians - Expectation of improvement or recovery - Physical activity - General health <u>Team-level:</u> - Received chiropractic care <u>Organizational-level:</u> - Year (2014, 2015, 2016)	An inverse relationship was identified between receiving chiropractic care and opioid prescribing (HR: 0.48, 99% CI: 0.29–0.77). After one year, 51% of patients who did not receive chiropractic care had been prescribed opioids compared to 29% of those who received chiropractic care. For patients who initiated chiropractic care early (within 30 days), the risk of starting an opioid prescription was lower than people who didn't receive chiropractic care (HR: 0.29, 99% CI: 0.13–0.68). At one year, 49% of patients without early chiropractic care received opioids, compared to only 18% of early chiropractic care recipients. Patients with more recent index visit dates were less likely to receive opioids (HR: 0.86, 99% CI: 0.76–0.97). Patients who received chiropractic care were less likely to fill an opioid prescription (IRR = 0.66; 95% CI, 0.52–0.83) or refill an opioid prescription (IRR = 0.27;
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Community With Limited Access to Health Care Services: A Description of Integration of Chiropractors as Service Providers (2017) Service evaluation					mean time between the completion of the pre and post-questionnaire was 10.2 +/- 8.2 weeks, ranging from 1 to 47 weeks.			Other healthcare utilization		95% CI, 0.17–0.42). At baseline, there was no significant difference in the odds of being prescribed a high opioid dose (≥ 50 mg MED) between chiropractic care recipients and non-recipients (OR = 0.61; 95% CI, 0.26–1.47). However, chiropractic recipients were significantly less likely to be prescribed higher opioid doses at 3 months (OR = 0.14; 95% CI, 0.04–0.47), 6 months (OR = 0.14; 95% CI, 0.05–0.40), 9 months (OR = 0.19; 95% CI, 0.07–0.57), and 12 months (OR = 0.22; 95% CI, 0.08–0.62). By the 12-month follow-up, 59% (29/49) of chiropractic recipients had discontinued opioid use compared to 31% (50/161) of non-recipients. Patients with a more recent index visit date had a lower rate of opioid refills (IRR = 0.82; 95% CI, 0.73–0.93) and were less likely to receive high-dose opioids at 3 months (OR = 0.73; 95% CI, 0.57–0.94) and 6 months (OR = 0.78; 95% CI, 0.62–0.99). A higher frequency of healthcare visits was associated with a greater likelihood of opioid refills (IRR = 1.06; 95% CI, 1.02–1.09) and
Emery P. (2022)										
Canada										

										higher opioid dosages at 3 months (OR = 1.11; 95% CI, 1.02–1.21), 6 months (OR = 1.09; 95% CI, 1.01–1.18), 9 months (OR = 1.10; 95% CI, 1.02–1.19), and 12 months (OR = 1.12; 95% CI, 1.03–1.21). EQ-5D scores indicated that two-thirds (67%) of patients showed improvement at follow-up, while less than one in ten (8.9%) had worsened. The remainder of patients either remained about the same (20%) or showed mixed improvement (4.4%). Analyzing the five EQ-5D domains, the most significant improvements (in decreasing order) were reported in self-care, anxiety/depression, usual activities, and mobility, with the smallest improvement seen in pain/discomfort Assuming primary care visits at Lang's CHC lasted 30 minutes and using an average wage of \$86.42 for medical services, the reduced total number of primary care visit hours for this patient cohort over 90 days would range between 23.4 and 71 hours. This represents a potential cost saving of between \$2,022.23 and \$6,135.82 for the
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										program. 77% of patients had not sought help from other practitioners for their back pain during this period. 82% of patients were either not taking any medication or had significantly reduced their medication usage for pain management. 93% patients expressed satisfaction with the treatment they received from the service, reporting they were either very satisfied or satisfied. For the Bournemouth Questionnaire, 63% of patients were classified as significantly improved. 74% of patients reported improvement on the Bothersomeness Scale, with no patients indicating that their back pain had worsened. For the Global Improvement Scale, very few patients reported being worse or experiencing no change in their back pain. 93% reported feeling better, although only 3 patients considered themselves
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										completely better.
Nurse practitioners, physician assistants, and physicians in community health centers, 2006 to 2010 (2015) Time spent with patients by physicians, nurse practitioners, and physician assistants in community health centers, 2006 to 2010 (2014) <i>Morgan P.</i> USA	Cross-sectional study (2015 and 2014) Morgan (2015): 62.5% Morgan (2014): 75%	Compare the amount of time spent with patients based on whether the provider was a physician, NP, or PA in CHCs.	Federally Qualified Health Center (FQHC) clinics that receive Section 330 grants under the Public Health Service Act. "Look-alike" health centers meeting the FQHCs requirements and federally qualified Indian health service clinics were also included.	FQHCs receiving funds from the Section 330 grant in the U.S. had 24,528 patient visits recorded from 2006 to 2010. All sampled providers were asked to complete a provider induction survey and patient visit forms for a randomly selected sample of up to 30 patient visits during a randomly chosen one-week period. The response rate for physicians was 85.5% and 86.6% for NP and PA. Sample size= 670 physicians 245 NP 103 PA 25 528 patients visit	Data source: the National Ambulatory Medical care survey community health centers stratum (2006 to 2010) Tool: physicians, NP and PA were asked to complete an induction survey and provide 30 patient visits forms for a randomly selected sample over a randomly selected one-week period. Surveys were sent in 2010. Timeline: 2006 to 2010	The provider type who saw the patient during the visit: physician, NP or PA <u>Team composition:</u> - Physician - NP - PA	Each type of provider was compared to the two others NP/PA compared to physician	Diagnosis code per visit: categorized into 7 groups (acute, chronic, dermatologic, mental health, pregnancy and prevent, ill-defined symptoms, vision and hearing) Number of visits per provider Time spent with patient Number of patients seen weekly Number of cumulated days spent with patients per clinical days	<u>Patient-level:</u> - Age - Sex - Race/ethnicity - Urban/rural location - Major reason for visit - Presence of chronic condition - Patient relation to the FQHC (new to the clinic, number of visits in the past 12 months, etc.) - Socio-economic status <u>Team-level:</u> % of patients seen before in this clinic (versus first-time visitors) % of patients that have been seen before in the last 12 months - Primary care provider status <u>Provider characteristics:</u> - Age - Sex - Location of FQHCs - Region (northeast, Midwest, southwest) - Race/ethnicity	CHC providers consisted of 69% physicians, 21% NPs, and 10% PAs. The distribution of visits by provider type was similar, with 65% of visits made by physicians, 23% by NPs, and 11% by PAs. Together, NPs and PAs accounted for over one-third of patient visits. The proportion of visits by patients in different age groups varied across provider types. A higher percentage of visits to NPs (48%) were made by young adults compared to visits to physicians (33%) or PAs (39%). Conversely, fewer visits to NPs were made by middle-aged and older adults, with physicians and PAs seeing similar proportions in these age groups. NPs had a larger proportion of visits with women (74%) compared to physicians (62%) or PAs (60%). Additionally, a greater proportion of visits to NPs and PAs (combined) were made by patients in rural CHCs (21%) compared to visits to physicians (9%; $p < 0.05$). Patient

										race/ethnicity was similar across provider types. A larger proportion of NP visits (42%) were for preventive care, compared to 29% of physician visits and 23% of PA visits. A smaller percentage of NP visits (17%) were for routine chronic care, compared to 26% for physicians and 25% for PAs ($p < 0.05$). PAs more frequently saw patients for new problems (45% vs. 38% for physicians and 36% for NPs). Fewer NP visits involved chronic disease (37% for NPs, 48% for physicians, 45% for PAs; $p < 0.05$), and more NP visits were for pregnancy or preventive care (34% for NPs vs. 22% for physicians, $p < 0.01$). There were no significant differences in whether the patient was new to the clinic across visits to physicians, NPs, and PAs. However, physicians saw a larger proportion of patients with the most visits to the CHC in the previous year compared to both NPs and PAs ($p < 0.01$). A larger percentage of physicians (76%) than NPs (59%) identified themselves as the primary care provider for patients they saw ($p <$
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										0.05). In urban CHC settings, physicians reported being the primary care provider for 18% more patients than NPs/PAs (76% vs. 57%), but in rural CHCs, both groups reported similar prevalence (around 80% of visits). During the reporting week, PAs spent slightly more time seeing patients, with an average of 3.8 days, compared to physicians (3.5 days) and NPs (3.4 days). When analyzing the total number of patients seen over the week, there was no statistical difference between provider types. However, after adjusting for the number of days spent in patient care, PAs saw fewer patients per clinical care day than physicians (14.8 patients per day for physicians vs. 12.9 for PAs, $p < 0.05$). The mean time spent with patients was 19.82 minutes for physicians, 19.96 minutes for NPs, and 20.15 minutes for PAs. These differences were minimal (within 20 seconds per visit) and were not statistically significant.
A Descriptive Analysis of a Community Clinic Providing Hepatitis C Treatment to Poor and	Cross-sectional	Examine the clinical characteristics of these patients, reports treatment completion and	Mercy Health Center (MHC) is a small nonprofit community clinic situated in northeast Georgia, United	Patients with confirmed hepatitis C condition confirmed by PCR testing. Out of 3,369 eligible patients, 60 were included in this	Data source: retrospective chart review Timeline: January 1 st 2008 to March 31 st	The HCV clinic at Mercy Health Center (MHC) employs a multidisciplinary care model for managing HCV, involving a team of professionals including a public health social worker, pharmacist, gastroenterologist,	Patient who did not receive care from the HCV clinic.	Completion of treatment with the HCV clinic SVR rates Estimated cost	Patient-level: ○ Age at initial visit ○ Sex ○ Race ○ Relationship status ○ Medical and	More than 1/3 patients received HCV treatment, which included all-oral interferon-free DAAs (12%), DAAs combined with pegylated interferon and ribavirin (46%), and

Uninsured Patients <i>Sims O (2018)</i> USA		sustained virologic response (SVR) rates, and estimates the cost of HCV medications provided free of charge to uninsured individuals.	States. MHC offers primary and specialty care services at no cost to patients who are uninsured, live within 150% of the federal poverty level (FPL), and reside within its six-county service area.	study. Sample size=69 participants	2015	nurse, nurse practitioner, psychologist, and dietitian. This treatment team meets weekly to assess patient readiness and eligibility for HCV treatment, as well as to prepare and submit applications to pharmaceutical patient assistance programs to secure free HCV medications for uninsured patients. If an application is denied, the public health social worker and nurse collaborate with the pharmacist and gastroenterologist to provide additional clinical documentation and medical justification for treatment. Before initiating therapy, patients attend personalized appointments to review laboratory results, understand the purpose and benefits of antiviral therapy, learn about medication adherence and proper storage and administration, and discuss potential side effects of the treatment. <u>Team composition:</u> - Public health social worker - Pharmacist - Gastroenterologist - Nurse - NP - Psychologist - Dietician		of medication coverage	mental health condition - Smoking status - History of alcohol and substance use - HCV exposure - HCV genotype - Liver histology - Biological markers of HCV <u>Team and organizational level:</u> none measured	pegylated interferon with ribavirin alone (42%). Among those treated, 81% successfully completed therapy, and 85% achieved sustained virologic response (SVR). The multidisciplinary HCV clinic facilitated the provision of HCV medications valued at an estimated \$1,447,163–\$1,457,129 at no cost to uninsured patients.
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Abbreviations: A1C: Glycated hemoglobin; AAPA: American Academy of Physician Associate; ACA: Affordable Care Act; ACEI: Angiotensin-converting enzyme inhibitor; ACS: American Community Service; ADA: American Diabetes Association; AL: Alabama; ARB: Angiotensin II receptor blocker; AHC: Affinia healthcare; AHRF: Area health resource file; AHW: Aboriginal health worker; AIU: Adoption implementation on upgrade; APC: Advanced practice clinician; APN: Advanced practice nurse; ARRA: American recovery and reinvestment act; BMI: Body mass index; BP: Blood pressure; BRFSS: Behavioral risk factor surveillance system; BQ: Bournemouth questionnaire; CCG: Community care group; CI: Confidence interval; CIP: Community investment program; CG-CAHPS: the Consumer Assessment of Healthcare Provider and System Clinician & Group; CHC: Community health centers; CHW: Community health worker; CMS: Center for Medicare and Medicaid service; CNS: Clinical nurse specialist; CO: Colorado; CSQ: Client satisfaction scale; CT: Connecticut; DAA: Direct oral antiviral; DC: District of Columbia; DEA: Data envelopment analysis; DE: Delaware; DFD: Depression free day; DMC: Depression care manager; EHR: Electronic health records; EQ-5: EuroQol 5-dimension; FIP: Federal internship program; FIT: Fecal immunochemical test; FOBI: Fecal occult blood test; FQHC: Federally qualified health center; FPL: Federal poverty level; FTE: Full-time equivalency; GP: General practitioner; HbA1C: Glycated hemoglobin; HCP: Healthcare provider; HCV: Hepatitis C virus; HCV RNA: Hepatitis C virus ribonucleic acid; HIV: Human immunodeficiency virus; HGM: Home glucose monitoring; HR: Hazard ratio; HRSA Health Resources Service Utilization; ICD-9: International Classification of Disease; IDS: Integrated data system; ITT: Incidence rate ratio; IRS: Internal revenue service; LCA: Latent class analysis; LCHC: Lincoln community health center; LDL: Low-density lipoprotein; MA: medical assistant; MCW: Manageable clinical workload; M: Mean; MA: Massachusetts; MCS: Mental composite score; MD: Maryland; MHC: Mercy Health center; MI: Michigan; MN: Minnesota; MO: Missouri; MoH: Ministry of Health; MSA: Metropolitan statistical area; MSK: Musculoskeletal; NA: Not applicable; NAP: New Access Point; ND: North Dakota; NDI: Neck disability index; NHSC: National health service corps; NP: Nurse practitioner; NV: Nevada; OR: Odd ratio; PA: Physician assistant; PACIC-II: Patient assessment of chronic illness care; PAID-5: Problem areas in diabetes; PBCC: Practice-based collaborative care; PC: Primary care; PCMH: Patient-centered medical home; PCP: Primary care provider; PDQ: Pain Disability Questionnaire; PGIC: Patient global impression of change; PH-9: Patient health questionnaire – 9; QALY: Quality adjusted life year; QRS: Quality of staff relationship; QWB: Quality of wellbeing scale; RCT: Randomized controlled trial REC: Regional educational center; RI: Rhode Island; RMQ: Roland-Morris Questionnaire SCL-20: Symptom checklist 20; SD: Standard deviation; SF-12: Short-Form 12; SOP: Scope of practice; SVR-12: Sustained virologic response; TBCC: Telemedicine-based collaborative care; THC: Teaching health center; TV: television; UAM: University of Arkansas medical science; UDS: Uniform Data Service; US: United States of America; VAS: Visual Analogue Scale; VI: Vermont; WES: Work environment scale-real; ZCAT: Zip code tabulation area

Appendix C. Risk of bias

Cross-sectional

Title, First author and Year of publication, Risk of bias score (% of "yes" out of all criteria)	Were the criteria for inclusion in the sample clearly defined?	Were the study subjects and the setting described in detail?	Was the exposure measured in a valid and reliable way?	Were objective, standard criteria used for measurement of the condition?	Were confounding factors identified?	Were strategies to deal with confounding factors stated?	Were the outcomes measured in a valid and reliable way?	Was appropriate statistical analysis used?
A Descriptive Analysis of a Community Clinic Providing Hepatitis C Treatment to	1	0	0	1	1	0	1	0

Poor and Uninsured Patients Sims O. 2018 50%								
An evaluative study of a community health service development Young K. (1994) 12.5%	0	0	0	0	0	1	0	0
Brokering Language and Culture Can Ad Hoc Interpreters Fill the Language Service Gap at Community Health Centers Larrison C. (2010) 62.5%	1	1	1	1	0	0	1	0
Changes in primary care access at community health centers between 2012-2013 and 2016 Saloner B. (2019) 100%	1	1	1	1	1	1	1	1
Chiropractic care of musculoskeletal disorder in a	1	1	1	1	1	1	1	1

unique population within Canadian community health center Garner M. (2007) 100%								
Clinic Workload, the Quality of Staff Relationships and Diabetes Management in Community Health Centers Catering to Latino and Chinese Patients Vargas Bustamante A. (2017) 75%	0	0	1	1	1	1	1	1
Community health center efficiency the impact of organization design and local context the case of Indonesia Miharti S. (2022) 75%	1	0	1	1	1	1	1	0
Community health centers employ diverse staffing patterns which can provide	1	0	1	1	1	1	1	1

productivity lessons for medical practices Ku L. (2015) 87.5%								
Community health center residency training improving staffing service quality Chen C. (2021) 87.5%	1	0	1	1	1	1	1	1
Does expanding state scope of practice for nurse practitioner and physician assistants increase primary care utilization in community health centers Park J. (2019) 87.5%	1	0	1	1	1	1	1	1
Does the patient-centered medical home model change staffing and utilization in the community health center Park J. (2018)	0	0	1	0	1	1	1	1

62.5%								
Do years of experience with electronic health records matter for productivity in community health center Frogner B. (2017)	0	0	1	1	1	1	1	1
75%								
Enhancing Staffing In Rural Community Health Centers Can Help Improve Behavioral Health Care Han X. (2019)	0	0	1	1	1	1	1	1
75%								
Management of Back Pain-related Disorders in a Community With Limited Access to Health Care Services A Description of Integration of Chiropractors as Service Providers Emary P. (2017)	1	1	1	1	0	0	1	0
62.5%								
Nurse practitioners,	0	1	1	1	0	0	1	1

physician assistants, and physician in community health center 2006-2010 Morgan P. (2015) 62.5%								
Organizational factors associated with disparities in cervical and colorectal cancer screening rates in community health centers Chuang E. (2019) 87.5%	1	0	1	1	1	1	1	1
Provision of telemedicine services by community health centers Shin P. (2014) 25%	0	0	0	0	0	0	1	1
Time spent with patients by physicians, nurse practitioners and physician assistant in community health centers 2006-2010 Morgan P. (2014)	1	0	1	1	1	1	0	1

75%								
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Cohort

Title, First author and Year of publication, Risk of bias score (% of "yes" out of all criteria)	Were the two groups similar and recruited from the same population?	Were the exposures measured similarly to assign people to both exposed and unexposed groups?	Was the exposure measured in a valid and reliable way?	Were confounding factors identified?	Were strategies to deal with confounding factors stated?	Were the groups/participants free of the outcome at the start of the study (or at the moment of exposure)?	Were the outcomes measured in a valid and reliable way?	Was the follow up time reported and sufficient to be long enough for outcomes to occur?	Was follow up complete, and if not, were the reasons to loss to follow up described and explored?	Were strategies to address incomplete follow up utilized?	Was appropriate statistical analysis used?
Aboriginal health workers and diabetes care in remote community health centers: a mixed method analysis Si D. (2006) 90.9%	1	1	1	1	1	1	1	1	1	0	1
Adoption of telepharmacy within a community health center A focus on clinical pharmacy services McCarthy C. (2021)	0	1	1	0	0	0	0	0	0	0	0

18.18%											
Association of Chiropractic Care With Receiving an Opioid Prescription for Noncancer Spinal Pain Within a Canadian Community Health Center: A Mixed Methods Analysis Emary P. (2022)	1	1	1	1	1	1	1	1	0	0	1
81.81%											
Cardiovascular care guideline implementatio n in community health centers in Oregon: a mixed- methods analysis of real-world barriers and challenges Gold R. (2017)	1	1	1	1	1	1	1	1	0	0	1
81.81%											
Improving Access to Hepatitis C Treatment Using Clinical Pharmacist Services in an Uninsured At- Risk	0	0	1	0	0	0	1	1	1	1	0

Population at a Community Health Center Fitzgerald L. (2021) 45.45%											
Influence of an interdisciplinary diabetes specialist team on short-term outcomes of diabetes at a community health center Katz G. (1998) 18.18%	0	0	1	0	0	0	0	1	0	0	0
Integrating a multidisciplinary pain team and chiropractic care in a community health center an observational study of managing chronic spinal pain Prater C. (2020) 81.81%	1	1	1	1	1	1	1	1	0	0	1
Optimal staffing in community health centers to improve quality of care	0	0	1	1	1	0	1	0	0	0	1

Luo Q. (2021) 45.45%											
Specialized Care Of Overweight Children In Community Health Centers Shin P. (2014) 18.18%	0	0	1	0	0	0	1	0	0	0	0
The association between chiropractic integration in an Ontario community health center and continued prescription opioid use for chronic non-cancer spinal pain a sequential explanatory mixed methods study Emary P. (2022) 81.81%	1	1	1	1	1	0	1	1	1	1	0
The role of the national health service corps clinicians in enhancing staffing and patient care capacity in community health centers	0	1	1	1	1	1	1	0	0	0	1

Han X. (2019)											
63.63%											

Economic Evaluation

Title, First author and Year of publication, Risk of bias score (% of "yes" out of all criteria)	Is there a well-defined question?	Is there a comprehensive description of alternatives?	Are all important and relevant costs and outcomes for each alternative identified?	Has clinical effectiveness been established?	Are costs and outcomes measured accurately?	Are costs and outcomes valued credibly?	Are costs and outcomes adjusted for differential timing?	Is there an incremental analysis of costs and consequences?	Were sensitivity analyses conducted to investigate uncertainty in estimates of cost or consequences?	Do study results include all issues of concern to users?	Are the results generalizable to the setting of interest in the review?
Chiropractic integration within a community health center a cost description and partial analysis of cost utility from the perspective of the institution Emery P. (2019) 72.72%	1	1	1	1	1	1	0	0	1	1	0
Cost effectiveness of on-site versus off site collaborative care for depression	1	1	1	1	1	1	0	1	1	1	1

in rural FQHCS Pyne J. (2015) 90.9%											
Diabetes management at community health centers examining associations with patient and regional characteristics efficiency and staffing patterns Thorsen M. (2020) 54.54%	1	0	1	1	1	1	0	0	0	0	1

Quasi-experimental

Title, First author and Year of publication, Risk of bias score (% of "yes" out of all criteria)	Is it clear in the study what is the "cause" and what is the "effect" (i.e. there is no confusion about which variable comes first)?	Was there a control group?	Were participants included in any comparisons similar?	Were the participants included in any comparisons receiving similar treatment/care, other than the exposure or intervention of interest?	Were there multiple measurements of the outcome, both pre and post the intervention/exposure?	Were the outcomes of participants included in any comparisons measured in the same way?	Were outcomes measured in a reliable way?	Was follow-up complete and if not, were differences between groups in terms of their follow-up adequately described and analyzed?	Was appropriate statistical analysis used?
A quasi-experiment	1	1	1	1	0	1	1	0	1

assessing the six-months effects of a nurse care coordination program on patient care experiences and clinician teamwork in community health centers Nembhard I. (2020) 77.7%												
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Randomized controlled trial

Title, First author and Year of publication, Risk of bias score (% of "yes" out of all criteria)	Was true randomization used for assignment of participants to treatment groups?	Was allocation to treatment groups concealed?	Were treatment groups similar at the baseline?	Were participants blind to treatment assignment?	Were those delivering the treatment blind to treatment assignment?	Were treatment groups treated identically other than the intervention of interest?	Were outcome assessors blind to treatment assignment?	Were outcomes measured in the same way for treatment groups?	Were outcomes measured in a reliable way?	Was follow up complete and if not, were differences between groups in terms of their follow up adequately described and analyzed?	Were participants analyzed in the groups to which they were randomized?	Was appropriate statistical analysis used?	Was the trial design appropriate and any deviations from the standard RCT design (individual randomization, parallel groups) accounted for in the conduct and analysis of the trial?
Practice-Based Versus	1	0	1	0	0	0	1	1	1	0	1	1	1

Telemedicine-Based Collaborative Care for Depression in Rural Federally Qualified Health Centers A Pragmatic Randomized Comparative Effectiveness Trial Fortney J. (2013) 61.53%													
The impact of integrating medical assistants and community health workers on diabetes care management in community health centers Rodriguez H. (2018) 38.46%	0	0	0	0	0	0	0	1	1	0	1	1	1